Embroidermodder

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1 Contributor Covenant Code of Conduct

1.1 Our Pledge

We as members, contributors, and leaders pledge to make participation in our community a harassment-free experience for everyone, regardless of age, body size, visible or invisible disability, ethnicity, sex characteristics, gender identity and expression, level of experience, education, socio-economic status, nationality, personal appearance, race, religion, or sexual identity and orientation.

We pledge to act and interact in ways that contribute to an open, welcoming, diverse, inclusive, and healthy community.

1.2 Our Standards

Examples of behavior that contributes to a positive environment for our community include:

- Demonstrating empathy and kindness toward other people
- Being respectful of differing opinions, viewpoints, and experiences
- Giving and gracefully accepting constructive feedback
- Accepting responsibility and apologizing to those affected by our mistakes, and learning from the experience
- Focusing on what is best not just for us as individuals, but for the overall community

Examples of unacceptable behavior include:

- The use of sexualized language or imagery, and sexual attention or advances of any kind
- Trolling, insulting or derogatory comments, and personal or political attacks
- Public or private harassment
- Publishing others' private information, such as a physical or email address, without their explicit permission
- Other conduct which could reasonably be considered inappropriate in a professional setting

1.3 Enforcement Responsibilities

Community leaders are responsible for clarifying and enforcing our standards of acceptable behavior and will take appropriate and fair corrective action in response to any behavior that they deem inappropriate, threatening, offensive, or harmful.

Community leaders have the right and responsibility to remove, edit, or reject comments, commits, code, wiki edits, issues, and other contributions that are not aligned to this Code of Conduct, and will communicate reasons for moderation decisions when appropriate.

1.4 Scope

This Code of Conduct applies within all community spaces, and also applies when an individual is officially representing the community in public spaces. Examples of representing our community include using an official e-mail address, posting via an official social media account, or acting as an appointed representative at an online or offline event.

1.5 Enforcement

Instances of abusive, harassing, or otherwise unacceptable behavior may be reported to the community leaders responsible for enforcement at embroidermodder@gmail.com. All complaints will be reviewed and investigated promptly and fairly.

All community leaders are obligated to respect the privacy and security of the reporter of any incident.

1.6 Enforcement Guidelines

Community leaders will follow these Community Impact Guidelines in determining the consequences for any action they deem in violation of this Code of Conduct:

1.6.1 1. Correction

Community Impact: Use of inappropriate language or other behavior deemed unprofessional or unwelcome in the community.

Consequence: A private, written warning from community leaders, providing clarity around the nature of the violation and an explanation of why the behavior was inappropriate. A public apology may be requested.

1.6.2 2. Warning

Community Impact: A violation through a single incident or series of actions.

Consequence: A warning with consequences for continued behavior. No interaction with the people involved, including unsolicited interaction with those enforcing the Code of Conduct, for a specified period of time. This includes avoiding interactions in community spaces as well as external channels like social media. Violating these terms may lead to a temporary or permanent ban.

1.6.3 3. Temporary Ban

Community Impact: A serious violation of community standards, including sustained inappropriate behavior.

Consequence: A temporary ban from any sort of interaction or public communication with the community for a specified period of time. No public or private interaction with the people involved, including unsolicited interaction with those enforcing the Code of Conduct, is allowed during this period. Violating these terms may lead to a permanent ban.

1.6.4 4. Permanent Ban

Community Impact: Demonstrating a pattern of violation of community standards, including sustained inappropriate behavior, harassment of an individual, or aggression toward or disparagement of classes of individuals.

Consequence: A permanent ban from any sort of public interaction within the community.

1.7 Attribution

This Code of Conduct is adapted from the Contributor Covenant, version 2.0, available at https://www.contributor-covenant.org/version/2/0/code_of_conduct.html.

Community Impact Guidelines were inspired by Mozilla's code of conduct enforcement ladder.

For answers to common questions about this code of conduct, see the FAQ at https://www.contributor-covenant.org/faq. Translations are available at https://www.contributor-covenant.org/translations.

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4 Source Code Pro

`Source Code Pro` is a set of OpenType fonts that have been designed to work well in user interface (UI) environments.

4.1 Open source files

The fonts' source files and build instructions are available in the `repository's master branch`.

4.2 Getting involved

`Open an issue` or send a suggestion to Source Code's designer `Paul D. Hunt`, for consideration.

4.3 Releases

- Latest release
- All releases

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- All releases

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10 Embroidermodder User Manual

Build with `$ bash build.sh`.

11 Test Designs

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An example of:

1. ZSK file format
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15 The Embroidermodder Reference Manual

Released under the terms of the GNU Free Documentation License (FDL) v1.3 see sections/fdl-1.3.tex.

15.1 Build

To build, run:

```
$ latexmk -pdf
```

to clean up files after the build process run:

```
$ latexmk -c
```

15.1.1 Dependencies on Ubuntu

On Ubuntu, this line should get the necessary packages (FIX THIS):

```
$ sudo apt install texlive texlive-latex-base latexmk
```

15.2 Experimental Build

https://pypi.org/project/mkdocs-bibtex/
https://github.com/orzih/mkdocs-with-pdf

16 Test Designs

The files contained within this folder should not be used unless you are a software developer testing libembroidery. They may not be valid files. Consider using known valid files. USE AT YOUR OWN RISK AND HAVE FUN.

17 zlib/libpng License

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18 What is Embroidermodder?

IN ALPHA: VERY EXPERIMENTAL

Embroidermodder is a free machine embroidery software program. The newest version, Embroidermodder 2 can:

- edit and create embroidery designs
- estimate the amount of thread and machine time needed to stitch a design
- convert embroidery files to a variety of formats
- upscale or downscale designs
- run on Windows, Mac and Linux

For more information, see our website.

18.1 Dependencies

To build Embroidermodder 2 from source you will need at least:

- The Embroidermodder 2 source code.
- Qt (version >= 6.0).
- OpenGL (version >= 2.0).

Optionally, you may want to add:

- KDE for thumbnailer.
- Valgrind for debugging.
- CMake (version >= 2.0) for build (see Building Without CMake below if this is an issue).

Ubuntu repository packages: The Qt, KDE and Valgrind build dependencies can be installed easily by opening a terminal and issuing this command:

```
sudo apt-get install git build-essential qt6-dev-tools libqt6-opengl-dev \
    valgrind cmake
```

Fedora repository packages: The Qt, KDE and Valgrind build dependencies can be installed easily by opening a terminal and issuing this command:

```
sudo yum install git gdb gcc-c++ qt-devel kdelibs-devel valgrind
```

18.1.1 Included

Some of the dependencies have been included and CMake will default to these versions.

- GLFW

18.2 Building

Various methods of building can be found in the project-files subfolder. For most builds, it is as simple as opening a terminal in the project folder and typing:

```
./needle -b
```

18.2.1 Building Without Bash

```
git clone https://github.com/embroidermodder/embroidermodder
cd embroidermodder
mkdir build
cd build
cmake ..
cmake --build .
```

Then run with

```
./embroidermodder2
```

18.2.2 Building Without Cmake

We're working on a one-liner for systems without cmake.

```
# FIXME:
$(CC) -o embroidermodder2 \
    src/*.c src/*/*.c src/*.h src/*/*.h \
    -lglfw -lGL -lm
```

18.2.3 Testing One-liner (Bash-only)

Sometimes an issue you're having is fixed in the current version: it is recommended that you use a fresh build to ensure that the problem is still present before reporting this. This one-liner helps with this: place it in your `.bashrc` for future use.

```
alias run_current_em2="rm -f build.sh && wget
      https://raw.githubusercontent.com/Embroidermodder/Embroidermodder/refs/heads/main/build.sh && bash
      build.sh --current"
```

Then, if you have issues, you can capture the terminal output like this.

```
run_current_em2 &> build.log
```

If there is a new issue arising from this test script itself, this is also valuable to create an issue for.

18.3 Help Files

A copy of the reference manual and user manual will be included in the source code in `docs/` folder. Currently these are works in progress and aren't usable. Some notes that may be helpful are in markdown files in this directory.

18.4 Sample Files

Various sample embroidery design files can be found in the `embroidermodder2/docs/samples` folder.

18.5 Install/Uninstall

Embroidermodder 2 is self contained, so it does not need to be installed. If you move the `embroidermodder2` executable to any other folder it should run exactly the same.

19 KDE4 Thumbnailer

This folder contains a thumbnailer for KDE4 and is intended to be used with KDE's file manager: Dolphin. It is responsible for creating the thumbnails you see when your mouse hovers over or clicks on an embroidery file.

20 Todo List

Global load_script (char ∗asset_dir, char ∗fname)

loading of nested structures like arrays of dicts etc.

21 Hierarchical Index

21.1 Class Hierarchy

This inheritance list is sorted roughly, but not completely, alphabetically:

Button_ 26

CommandData_ 27

DocumentData_ 28

GroupBoxData_ 29

LayerData_ 30

ObjectCore_ 33
QApplication

Application 25
QDialog

LayerManager 30
QDockWidget

PropertyEditor 34

UndoEditor 37
QGraphicsPathItem

Object 32
QGraphicsView

Document 27
QLineEdit

CmdPromptInput 26
QMainWindow

MainWindow 30
QMdiArea

MdiArea 31
QMdiSubWindow

MdiWindow 32
QRubberBand

SelectBox 34
QUndoCommand

UndoableCommand 35

Setting_ 35

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ThumbCreator

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Translation_ 35

UndoData_ 36

ViewData_ 37

Widget_ 38

WidgetData_ 38

WidgetDesc_ 39

22 Data Structure Index

22.1 Data Structures

Here are the data structures with brief descriptions:

Application 25

Button_ 26

CmdPromptInput 26

CommandData_ 27

Document 27

DocumentData_ 28

EmbroideryThumbnailer 29

GroupBoxData_ 29

LayerData_ 30

LayerManager 30

MainWindow 30

MdiArea 31

MdiWindow 32

Object 32

ObjectCore_ 33

PropertyEditor 34

SelectBox 34

Setting_ 35

Tab_ 35

Translation_ 35

UndoableCommand 35

UndoData_ 36

UndoEditor 37

ViewData_ 37

Widget_ 38

WidgetData_ 38

WidgetDesc_ 39

23 File Index

23.1 File List

Here is a list of all documented files with brief descriptions:

src/core.h 39

src/interface.c 62

src/objects.c 67

src/qt.h 69

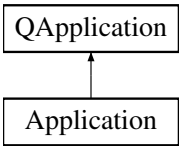
src/state.c 73

thumbnailer/libembroidery-thumbnailer-kde4.h 91

24 Data Structure Documentation

24.1 Application Class Reference

Inheritance diagram for Application:



Public Member Functions

- **Application** (int argc, char ∗∗argv)

Protected Member Functions

- virtual bool **event** (QEvent ∗e)

The documentation for this class was generated from the following files:

- src/qt.h
- src/commands.cpp

24.2 Button_ Struct Reference

Data Fields

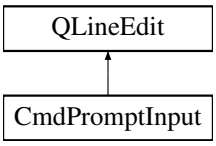
- EmbRect **rect**
- EmbColor **color**
- EmbColor **highlight_color**
- char ∗ **text**
- char ∗ **font**
- EmbColor **text_color**
- int **state**

The documentation for this struct was generated from the following file:

- src/core.h

24.3 CmdPromptInput Class Reference

Inheritance diagram for CmdPromptInput:



Public Slots

- void **checkSelection** ()
- void **updateCurrentText** (QString txt)
- void **checkEditedText** (QString txt)
- void **checkChangedText** (QString txt)
- void **checkCursorPosition** (int oldpos, int newpos)

Public Member Functions

- **CmdPromptInput** (QWidget ∗parent=0)

Protected Member Functions

- void **contextMenuEvent** (QContextMenuEvent ∗event)
- bool **eventFilter** (QObject ∗obj, QEvent ∗event)

The documentation for this class was generated from the following files:

- src/qt.h
- src/commands.cpp

24.4 CommandData_ Struct Reference

Data Fields

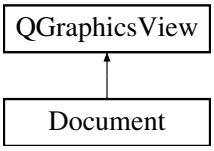
- char ∗ **command**
- char ∗ **arguments**
- char ∗ **icon**
- char ∗ **tooltip**
- char ∗ **statustip**
- char ∗ **alias**
- char ∗ **shortcut**
- int32_t **flags**
- Command **action**

The documentation for this struct was generated from the following file:

- src/core.h

24.5 Document Class Reference

Inheritance diagram for Document:



Public Member Functions

- **Document** (QGraphicsScene ∗theScene, QWidget ∗parent)

Data Fields

- DocumentData ∗ **data**
- QVector< int64_t > **hashDeletedObjects**
- QPainterPath **gridPath**
- QPainterPath **originPath**
- QPainter ∗ **scene_painter**
- Object ∗ **gripBaseObj**
- QGraphicsScene ∗ **gscene**
- QUndoStack ∗ **undoStack**
- SelectBox ∗ **selectBox**
- QGraphicsItemGroup ∗ **pasteObjectItemGroup**

Protected Member Functions

- void **mouseDoubleClickEvent** (QMouseEvent ∗event)
- void **mousePressEvent** (QMouseEvent ∗event)
- void **mouseMoveEvent** (QMouseEvent ∗event)
- void **mouseReleaseEvent** (QMouseEvent ∗event)
- void **wheelEvent** (QWheelEvent ∗event)
- void **contextMenuEvent** (QContextMenuEvent ∗event)
- void **drawBackground** (QPainter ∗painter, const QRectF &rect)
- void **draw_rulers** (QPainter ∗painter, const QRectF &rect)
- void **drawForeground** (QPainter ∗painter, const QRectF &rect)
- void **enterEvent** (QEvent ∗event)

The documentation for this class was generated from the following files:

- src/qt.h
- src/commands.cpp

24.6 DocumentData_ Struct Reference

Data Fields

- int32_t **id**
- EmbPattern ∗ **pattern**
- ScriptValue **attributes** [N_VIEW_ATTR]
- UndoData **undo_stack** [MAX_UNDO]
- EmbIdList ∗ **selectedItems**
- EmbIdList ∗ **rubberRoomList**
- EmbIdList ∗ **previewObjectList**
- EmbIdList ∗ **spareRubberList**
- EmbIdList ∗ **cutCopyObjectList**
- EmbIdList ∗ **previewObjectItemGroup**
- EmbIdList ∗ **hashDeletedObjects**
- EmbIdList ∗ **pasteObjectItemGroup**
- EmbVectorList ∗ **selectedGripPoints**
- bool **qSnapActive**
- bool **qSnapToggle**
- EmbVector **previewPoint**
- EmbReal **previewData**
- int **previewMode**
- EmbVector **viewMousePoint**
- EmbVector **sceneMousePoint**
- EmbVector **sceneQSnapPoint**
- uint8_t **qsnapLocatorSize**
- uint32_t **qsnapLocatorColor**
- uint8_t **qsnapApertureSize**
- uint32_t **gripColorCool**
- uint32_t **gripColorHot**
- uint32_t **gridColor**
- uint8_t **gripSize**
- uint8_t **pickBoxSize**
- uint32_t **crosshairColor**
- uint32_t **crosshairSize**
- EmbVector **scenePressPoint**
- EmbVector **pressPoint**
- EmbVector **sceneMovePoint**
- EmbVector **movePoint**

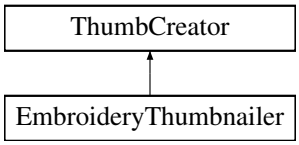
- EmbVector **sceneReleasePoint**
- EmbVector **releasePoint**
- EmbVector **sceneGripPoint**
- EmbVector **cutCopyMousePoint**
- EmbVector **pasteDelta**
- int **panDistance**
- int **panStartX**
- int **panStartY**
- uint32_t **rulerColor**
- bool **rulerMetric**
- uint8_t **rulerPixelSize**
- uint32_t **backgroundColor**
- bool **fileWasLoaded**
- int **myIndex**
- char **curFile** [MAX_STRING_LENGTH]
- char **curLayer** [MAX_STRING_LENGTH]
- uint32_t **curColor**
- char **curLineType** [MAX_STRING_LENGTH]
- char **curLineWeight** [MAX_STRING_LENGTH]

The documentation for this struct was generated from the following file:

- src/core.h

24.7 EmbroideryThumbnailer Class Reference

Inheritance diagram for EmbroideryThumbnailer:



Public Member Functions

- virtual bool **create** (const QString &path, int w, int h, QImage &img)

The documentation for this class was generated from the following files:

- thumbnailer/libembroidery-thumbnailer-kde4.h
- thumbnailer/libembroidery-thumbnailer-kde4.cpp

24.8 GroupBoxData_ Struct Reference

Data Fields

- int32_t **id**
- char ∗ **key**
- char ∗ **label**
- int **rows**
- char ∗ **data** [100]
- int **object**

The documentation for this struct was generated from the following file:

- src/core.h

24.9 LayerData_ Struct Reference

Data Fields

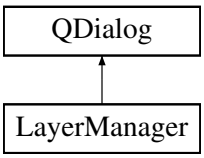
- char **name** [200]
- bool **visible**
- bool **frozen**
- EmbReal **zValue**
- int32_t **color**
- char ∗ **lineType**
- char ∗ **lineWeight**
- bool **print**

The documentation for this struct was generated from the following file:

- src/core.h

24.10 LayerManager Class Reference

Inheritance diagram for LayerManager:



Public Member Functions

- **LayerManager** (QWidget ∗parent=0)
- void **addLayer** (char ∗name, bool visible, bool frozen, EmbReal zValue, QRgb color, char ∗lineType, char ∗lineWeight, bool print)

Data Fields

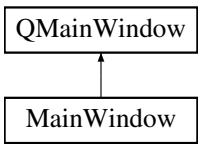
- LayerData **data** [MAX_LAYERS]
- QStandardItemModel ∗ **layerModel**
- QSortFilterProxyModel ∗ **layerModelSorted**
- QTreeView ∗ **treeView**

The documentation for this class was generated from the following files:

- src/qt.h
- src/commands.cpp

24.11 MainWindow Class Reference

Inheritance diagram for MainWindow:



Public Slots

- void **recentMenuAboutToShow** ()
- void **windowMenuActivated** (bool checked)
- void **closeToolBar** (QAction ∗)
- void **floatingChangedToolBar** (bool)
- void **runCommand** ()
- void **colorSelectorIndexChanged** (int index)

Public Member Functions

- void **add_toolbar_to_window** (Qt::ToolBarArea area, int data[ ])

Protected Member Functions

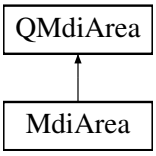
- virtual void **resizeEvent** (QResizeEvent ∗)
- void **closeEvent** (QCloseEvent ∗event)

The documentation for this class was generated from the following files:

- src/qt.h
- src/commands.cpp

24.12 MdiArea Class Reference

Inheritance diagram for MdiArea:



Public Slots

- void **cascade** ()
- void **tile** ()

Public Member Functions

- **MdiArea** (QWidget ∗parent=0)
- void **forceRepaint** ()

Data Fields

- bool **useLogo**
- bool **useTexture**
- bool **useColor**
- QPixmap **bgLogo**
- QPixmap **bgTexture**
- QColor **bgColor**

Protected Member Functions

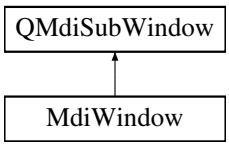
- virtual void **mouseDoubleClickEvent** (QMouseEvent ∗e)
- virtual void **paintEvent** (QPaintEvent ∗e)

The documentation for this class was generated from the following files:

- src/qt.h
- src/commands.cpp

24.13 MdiWindow Class Reference

Inheritance diagram for MdiWindow:



Public Slots

- void **closeEvent** (QCloseEvent ∗e)
- void **onWindowActivated** ()
- void **print** ()
- void **saveBMC** ()

Signals

- void **sendCloseMdiWin** (MdiWindow ∗)

Public Member Functions

- **MdiWindow** (const int theIndex, QMdiArea ∗parent, Qt::WindowFlags wflags)
- virtual QSize **sizeHint** () const
- QString **getShortCurrentFile** ()
- bool **loadFile** (char ∗fileName)

Data Fields

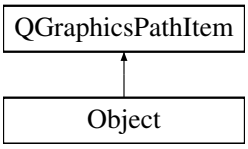
- QMdiArea ∗ **mdiArea**
- QGraphicsScene ∗ **gscene**
- int32_t **doc_index**
- QPrinter **printer**

The documentation for this class was generated from the following files:

- src/qt.h
- src/commands.cpp

24.14 Object Class Reference

Inheritance diagram for Object:



Public Member Functions

- **Object** (int type_, QRgb rgb, Qt::PenStyle lineType, QGraphicsItem ∗item=0)
- void **gripEdit** (EmbVector before, EmbVector after)
- QPainterPath **shape** () const
- void **drawRubberLine** (QLineF rubLine, QPainter ∗painter=0, const char ∗colorFromScene=0)
- void **setObjectSize** (double width, EmbReal height)

- QPainterPath **objectCopyPath** () const
- QPainterPath **objectSavePath** () const
- QList< QPainterPath > **objectSavePathList** () const
- QList< QPainterPath > **subPathList** () const

Data Fields

- ObjectCore * **core**
- QGraphicsPathItem **path_**
- QPen **objPen**
- QPen **lwtPen**
- QLineF **objLine**
- QPainter * **obj_painter**
- QPainterPath **textPath**
- QPainterPath **lineStylePath**
- QPainterPath **arrowStylePath**
- QPainterPath **normalPath**

Protected Member Functions

- void **paint** (QPainter *, const QStyleOptionGraphicsItem *, QWidget *)

The documentation for this class was generated from the following files:

- src/qt.h
- src/commands.cpp

24.15 ObjectCore_ Struct Reference

Data Fields

- int32_t **doc**
- EmbGeometry * **geometry**
- LabelledVector * **rubber_points**
- int **n_rubber_points**
- StringMap * **rubber_texts**
- int **n_rubber_texts**
- EmbVector **position**
- EmbReal **scale**
- EmbReal **rotation**
- uint32_t **rgb**
- char **OBJ_NAME** [MAX_STRING_LENGTH]
- char **text** [MAX_STRING_LENGTH]
- char **textFont** [MAX_STRING_LENGTH]
- char **textJustify** [MAX_STRING_LENGTH]
- EmbReal **textSize**
- bool **textBold**
- bool **textItalic**
- bool **textUnderline**
- bool **textStrikeOut**
- bool **textOverline**
- bool **textBackward**
- bool **textUpsideDown**
- int **rubber_mode**
- uint32_t **objID**
- bool **curved**
- bool **filled**

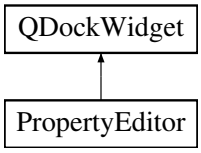
- EmbReal **arrowStyleAngle**
- EmbReal **arrowStyleLength**
- EmbReal **lineStyleAngle**
- EmbReal **lineStyleLength**
- int **gripIndex**

The documentation for this struct was generated from the following file:

- src/core.h

24.16 PropertyEditor Class Reference

Inheritance diagram for PropertyEditor:



Public Slots

- void **setSelectedItems** (QList< QGraphicsItem ∗ > itemList)
- void **update_pick_add_modeButton** (bool pickAddMode)

Signals

- void **pick_add_mode_toggled** ()

Public Member Functions

- **PropertyEditor** (QString iconDirectory="", bool pickAddMode=true, QWidget ∗widgetToFocus=0, QWidget ∗parent=0)
- void **togglePickAddMode** ()

Protected Member Functions

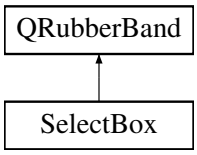
- bool **eventFilter** (QObject ∗obj, QEvent ∗event)

The documentation for this class was generated from the following files:

- src/qt.h
- src/commands.cpp

24.17 SelectBox Class Reference

Inheritance diagram for SelectBox:



Public Slots

- void **setDirection** (int dir)
- void **setColors** (const QColor &colorL, const QColor &fillL, const QColor &colorR, const QColor &fillR, int newAlpha)

Public Member Functions

- **SelectBox** (Shape s, QWidget ∗parent=0)

Protected Member Functions

- void **paintEvent** (QPaintEvent ∗)

The documentation for this class was generated from the following files:

- src/qt.h
- src/commands.cpp

24.18 Setting_ Struct Reference

Data Fields

- char **key** [MAX_STRING_LENGTH]
- ScriptValue **setting**
- ScriptValue **dialog**
- ScriptValue **accept**
- ScriptValue **preview**

The documentation for this struct was generated from the following file:

- src/core.h

24.19 Tab_ Struct Reference

Data Fields

- int **state**

The documentation for this struct was generated from the following file:

- src/core.h

24.20 Translation_ Struct Reference

Data Fields

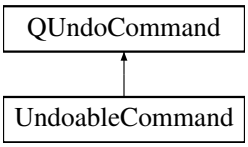
- char ∗ **ar**
- char ∗ **en**
- char ∗ **es**
- char ∗ **fr**
- char ∗ **ru**
- char ∗ **zh**
- int32_t **status**

The documentation for this struct was generated from the following file:

- src/core.h

24.21 UndoableCommand Class Reference

Inheritance diagram for UndoableCommand:



Public Member Functions

- **UndoableCommand** (int type_, QString text, Object ∗obj, int32_t v, QUndoCommand ∗parent=0)
- **UndoableCommand** (int type_, EmbVector delta, QString text, Object ∗obj, int32_t v, QUndoCommand ∗parent=0)
- **UndoableCommand** (int type_, EmbVector pivot, EmbReal rotAngle, QString text, Object ∗obj, int32_t v, QUndoCommand ∗parent=0)
- **UndoableCommand** (int type_, QString type, int32_t v, QUndoCommand ∗parent=0)
- **UndoableCommand** (int type_, EmbVector start, EmbVector end, QString text, Object ∗obj, int32_t v, QUndoCommand ∗parent=0)
- void **undo** ()
- void **redo** ()
- void **rotate** (double x, EmbReal y, EmbReal rot)
- int **id** () const
- bool **mergeWith** (const QUndoCommand ∗command)
- void **mirror** ()

Data Fields

- UndoData **data**
- Object ∗ **object**
- QTransform **toTransform**
- QTransform **fromTransform**

The documentation for this class was generated from the following files:

- src/qt.h
- src/commands.cpp

24.22 UndoData_ Struct Reference

Data Fields

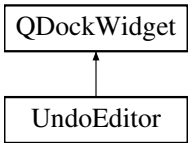
- char **label** [MAX_STRING_LENGTH]
- int **type**
- int32_t **doc**
- EmbVector **delta**
- EmbVector **pivot**
- EmbReal **angle**
- EmbReal **factor**
- char **navType** [MAX_STRING_LENGTH]
- EmbVector **fromCenter**
- EmbVector **toCenter**
- bool **done**
- EmbVector **before**
- EmbVector **after**
- EmbLine **mirrorLine**
- uint32_t **obj**

The documentation for this struct was generated from the following file:

- src/core.h

24.23 UndoEditor Class Reference

Inheritance diagram for UndoEditor:



Public Slots

- void **undo** ()
- void **redo** ()
- void **updateCleanIcon** (bool opened)

Public Member Functions

- **UndoEditor** (QString iconDirectory="", QWidget ∗widgetToFocus=0, QWidget ∗parent=0)
- void **addStack** (QUndoStack ∗stack)
- bool **canUndo** () const
- bool **canRedo** () const
- QString **undoText** () const
- QString **redoText** () const

The documentation for this class was generated from the following files:

- src/qt.h
- src/commands.cpp

24.24 ViewData_ Struct Reference

```
#include <core.h>
```

Data Fields

- int32_t **doc**
- double little
- double medium
- EmbLine ∗ lines
- int **n_lines**
- EmbGeometry ∗ **geometry**
- int **n_geometry**
- int **width**
- int **height**
- int **unit**
- bool **feet**
- double **fraction**
- int **xStart**
- int **yStart**

24.24.1 Detailed Description

Data necessary to create a view of the EmbPattern (pointer to this pattern is available via the doc index).

24.24.2 Field Documentation

24.24.2.1 lines `EmbLine* ViewData_::lines`

The size of the medium ticks on the rulers.

24.24.2.2 little `double ViewData_::little`

Index to the document in the list of open files.

24.24.2.3 medium `double ViewData_::medium`

The size of the small ticks on the rulers.

The documentation for this struct was generated from the following file:

- src/core.h

24.25 Widget_ Struct Reference

`#include <core.h>`

Data Fields

- EmbString key
- void * widget

24.25.1 Detailed Description

Generic widget pointer for a widget map.

24.25.2 Field Documentation

24.25.2.1 key `EmbString Widget_::key`

Used for identifying the element

24.25.2.2 widget `void* Widget_::widget`

A void pointer to represent a structure not in the C part of the source code.

The documentation for this struct was generated from the following file:

- src/core.h

24.26 WidgetData_ Struct Reference

`#include <core.h>`

Data Fields

- char * **key**
- char * label
- int id
- int type

24.26.1 Detailed Description

Generic widget data.

24.26.2 Field Documentation

24.26.2.1 id `int WidgetData_::id`

The label presented to the user in overlays.

24.26.2.2 label `char* WidgetData_::label`

The key used by Qt to deal with signals and slots.

24.26.2.3 type `int WidgetData_::type`

Used for identifying the element, if less than 0 then this is the terminator symbol for the list.

The documentation for this struct was generated from the following file:

- src/core.h

24.27 WidgetDesc_ Struct Reference

Data Fields

- int **id**
- char ∗ **label**
- char ∗ **key**
- char ∗ **icon**
- char ∗ **value**
- char ∗ **editor_data**
- int **type**
- int **n_leaves**

The documentation for this struct was generated from the following file:

- src/core.h

25 File Documentation

25.1 core.h

```
/*
* \file core.h
* \brief The C compatible parts of the header.  Generally the non-Qt parts.
*
* Embroidermodder 2.
*
* Copyright 2011-2025 The Embroidermodder Team
* Embroidermodder 2 is Open Source Software, see LICENSE.md for licensing terms.
*
* Read the reference manual (https://www.libembroidery.org/downloads/emrm.pdf)
* for advice on altering this file.
*
* C Core
*/

#ifndef EMBROIDERMODDER_C_CORE_HEADER
#define EMBROIDERMODDER_C_CORE_HEADER

#ifdef __cplusplus
extern "C" {
#endif

#include <inttypes.h>
#include <stdlib.h>
#include <stdbool.h>
#include <stdarg.h>

#include "embroidery.h"

/*
* Integer Constants
*/
#define MAX_ARGS                      10
#define MAX_FILES                     30
#define MAX_STRING_LENGTH            200
#define MAX_COMMANDS                 300
#define MAX_COMMAND_LENGTH            30
#define MAX_MENU_LENGTH               30
#define MAX_TOOLBAR_LENGTH            30
#define MAX_ALIASES                  500
#define MAX_STATE_VARIABLES          500
#define MAX_STATE_STRING_LENGTH      200
#define MAX_LABEL_LENGTH              50
```

```
#define MAX_UNDO                    100
#define MAX_ACTIONS                1000
#define MAX_OBJECTS               20000
#define MAX_WIDGETS                1000
#define MAX_EDITORS                1000
#define MAX_LONG_STRING            1000
#define MAX_TABLE_LENGTH            500
#define MAX_OPEN_FILES              100

#define NUMBINS                      10

#define WIDGET_COMBOBOX               0
#define WIDGET_LINEEDIT               1
#define WIDGET_GROUP_BOX              2
#define WIDGET_CHECKBOX               3
#define WIDGET_SPINBOX                4
#define WIDGET_LABEL                  5
#define WIDGET_UNKNOWN                6

/* main() is run every time the command is started.
* Use it to reset variables so they are ready to go.
*/
#define CONTEXT_MAIN                  0
/* click() is run only for left clicks.  Middle clicks are used for
* panning.  Right clicks bring up the context menu.
*/
#define CONTEXT_CLICK                 1
/* move() is optional.  It is run only after enableMoveRapidFire() is
* called.  It will be called every time the mouse moves until
* disableMoveRapidFire() is called.
*/
#define CONTEXT_MOVE                  2
/* context() is run when a context menu entry is chosen.  */
#define CONTEXT_CONTEXT               3
/* prompt() is run when Enter is pressed.
* appendPromptHistory is automatically called before prompt()
* is called so calling it is only needed for erroneous input.
* Any text in the command prompt is sent as an uppercase string.
*/
#define CONTEXT_PROMPT                4

/* Command requirements.  */
#define CONTEXT_FREE               0x00
#define REQUIRED_VIEW              0x01
#define REQUIRED_SCENE             0x02
#define REQUIRED_UNDO              0x04
#define DONT_INITIALIZE            0x08
#define CLEAR_SELECTION            0x10
#define DONT_END_COMMAND           0x20

/* Key Sequence */
#define CUT_SEQUENCE                  0
#define COPY_SEQUENCE                 1
#define PASTE_SEQUENCE                2
#define SELECT_ALL_SEQUENCE           3
#define UNDO_SEQUENCE                 4
#define REDO_SEQUENCE                  5
#define DELETE_KEY                     6
#define TAB_KEY                        7
#define ESCAPE_KEY                     8
#define UP_KEY                         9
#define DOWN_KEY                      10
#define SHIFT_KEY                     11
#define F1_KEY                        12
#define F2_KEY                        13
#define F3_KEY                        14
#define F4_KEY                        15
#define F5_KEY                        16
#define F6_KEY                        17
#define F7_KEY                        18
#define F8_KEY                        19
#define F9_KEY                        20
#define F10_KEY                       21
#define F11_KEY                       22
#define F12_KEY                       23
#define N_KEY_SEQUENCES               24

/* Command Actions */
#define ACTION_NULL                    0
#define ACTION_DO_NOTHING              1

#define ACTION_NEW                     2
#define ACTION_OPEN                    3
#define ACTION_SAVE                    4
#define ACTION_SAVE_AS                 5
#define ACTION_PRINT                   6
#define ACTION_DESIGN_DETAILS          7
```

```
#define ACTION_EXIT                    8
#define ACTION_CUT                     9
#define ACTION_COPY                   10
#define ACTION_COPY_SELECTED          11
#define ACTION_PASTE                  12
#define ACTION_PASTE_SELECTED         13

#define ACTION_UNDO                   14
#define ACTION_REDO                   15

/* Window Menu */
#define ACTION_WINDOW_CLOSE           16
#define ACTION_WINDOW_CLOSE_ALL       17
#define ACTION_WINDOW_CASCADE         18
#define ACTION_WINDOW_TILE            19
#define ACTION_WINDOW_NEXT            20
#define ACTION_WINDOW_PREVIOUS        21

/* Help Menu */
#define ACTION_HELP                   22
#define ACTION_CHANGELOG              23
#define ACTION_TIP_OF_THE_DAY         24
#define ACTION_ABOUT                  25
#define ACTION_WHATS_THIS             26

/* Icons */
#define ACTION_ICON_16                27
#define ACTION_ICON_24                28
#define ACTION_ICON_32                29
#define ACTION_ICON_48                30
#define ACTION_ICON_64                31
#define ACTION_ICON_128               32

#define ACTION_SETTINGS_DIALOG        33

/* Layer ToolBar */
#define ACTION_MAKE_LAYER_CURRENT     34
#define ACTION_LAYERS                 35
#define ACTION_LAYER_SELECTOR         36
#define ACTION_LAYER_PREVIOUS         37
#define ACTION_COLOR_SELECTOR         38
#define ACTION_LINE_TYPE_SELECTOR     39
#define ACTION_LINE_WEIGHT_SELECTOR   40
#define ACTION_HIDE_ALL_LAYERS        41
#define ACTION_SHOW_ALL_LAYERS        42
#define ACTION_FREEZE_ALL_LAYERS      43
#define ACTION_THAW_ALL_LAYERS        44
#define ACTION_LOCK_ALL_LAYERS        45
#define ACTION_UNLOCK_ALL_LAYERS      46

/* Text ToolBar */
#define ACTION_TEXT_BOLD              47
#define ACTION_TEXT_ITALIC            48
#define ACTION_TEXT_UNDERLINE         49
#define ACTION_TEXT_STRIKEOUT         50
#define ACTION_TEXT_OVERLINE          51

/* Zoom ToolBar */
#define ACTION_ZOOM_REAL_TIME         52
#define ACTION_ZOOM_PREVIOUS          53
#define ACTION_ZOOM_WINDOW            54
#define ACTION_ZOOM_DYNAMIC           55
#define ACTION_ZOOM_SCALE             56
#define ACTION_ZOOM_CENTER            57
#define ACTION_ZOOM_IN                58
#define ACTION_ZOOM_OUT               59
#define ACTION_ZOOM_SELECTED          60
#define ACTION_ZOOM_ALL               61
#define ACTION_ZOOM_EXTENTS           62

/* Pan SubMenu */
#define ACTION_PAN_REAL_TIME          63
#define ACTION_PAN_POINT              64
#define ACTION_PAN_LEFT               65
#define ACTION_PAN_RIGHT              66
#define ACTION_PAN_UP                 67
#define ACTION_PAN_DOWN               68

#define ACTION_DAY                    69
#define ACTION_NIGHT                  70

/* Prompt.  */
#define ACTION_ALERT                  71
#define ACTION_GET                    72
#define ACTION_SET                    73

/* Selection management.  */
```

```
#define ACTION_CLEAR                 74

/* TODO: ACTION_spellcheck, */
/* TODO: ACTION_quickselect, */

/* Geometry */
#define ACTION_ANGLE                 75
#define ACTION_CIRCLE                76
#define ACTION_DEBUG                 77
#define ACTION_DIM_LEADER            78
#define ACTION_DISABLE               79
#define ACTION_DISTANCE              80
#define ACTION_DOLPHIN               81
#define ACTION_ELLIPSE               82
#define ACTION_ENABLE                83
#define ACTION_ERASE                 84
#define ACTION_ERROR                 85
#define ACTION_HEART                 86
#define ACTION_LINE                  87
#define ACTION_LOCATE_POINT          88
#define ACTION_MIRROR_SELECTED       89
#define ACTION_MOVE                  90
#define ACTION_MOVE_SELECTED         91
#define ACTION_PATH                  92
#define ACTION_PLATFORM              93
#define ACTION_POINT                 94
#define ACTION_POLYGON               95
#define ACTION_POLYLINE              96
#define ACTION_PREVIEW_OFF           97
#define ACTION_PREVIEW_ON            98
#define ACTION_QUICKLEADER           99
#define ACTION_RECTANGLE            100
#define ACTION_RGB                  101
#define ACTION_ROTATE               102
#define ACTION_SCALE                104
#define ACTION_SCALE_SELECTED       105
#define ACTION_SELECT_ALL           106
#define ACTION_SINGLE_LINE_TEXT     107
#define ACTION_SNOWFLAKE            108
#define ACTION_STAR                 109
#define ACTION_SYSWINDOWS           110
#define ACTION_TODO                 111
#define ACTION_VULCANIZE            112

#define ACTION_ADD                  113
#define ACTION_DELETE               114
#define ACTION_GRIP_EDIT            115
#define ACTION_NAV                  116
#define ACTION_MIRROR               117

#define ACTION_TEST                 118

/* Simulation controls */
#define ACTION_SIMULATE             119
#define ACTION_PLAY                 120
#define ACTION_STOP                 121
#define ACTION_PAUSE                122
#define ACTION_FAST_FORWARD         123
#define ACTION_REWIND               124
#define ACTION_EXPORT_VIDEO         125

/* Generators */
#define ACTION_QR                   126
#define ACTION_LETTERING            127
#define ACTION_PATTERN              128
#define ACTION_DESIGN               129

#define N_ACTIONS                   130

/* UI Modes */
#define CIRCLE_1P_RAD_ONE             0
#define CIRCLE_1P_RAD_TWO             1
#define CIRCLE_1P_DIA_ONE             2
#define CIRCLE_1P_DIA_TWO             3
#define CIRCLE_2P                     4
#define CIRCLE_3P                     5
#define CIRCLE_TTR                    6
#define CIRCLE_TTT                    7

#define ELLIPSE_MAJDIA_MINRAD_ONE     0
#define ELLIPSE_MAJDIA_MINRAD_TWO     1
#define ELLIPSE_MAJDIA_MINRAD_THREE   2
#define ELLIPSE_MAJRAD_MINRAD_ONE     3
#define ELLIPSE_MAJRAD_MINRAD_TWO     4
#define ELLIPSE_MAJRAD_MINRAD_THREE   5
#define ELLIPSE_ROTATION              6
#define ELLIPSE_NEUTRAL               7
```

```

#define POLYGON_NUM_SIDES              0
#define POLYGON_CENTER_PT              1
#define POLYGON_POLYTYPE               2
#define POLYGON_INSCRIBE               3
#define POLYGON_CIRCUMSCRIBE           4
#define POLYGON_DISTANCE               5
#define POLYGON_SIDE_LEN               6

#define TEXTSINGLE_JUSTIFY             0
#define TEXTSINGLE_SETFONT             1
#define TEXTSINGLE_SETGEOM             2
#define TEXTSINGLE_RAPID               3

/* NOTE: Allow this enum to evaluate false */
#define ARROW_STYLE_NO_ARROW           0
#define ARROW_STYLE_OPEN               1
#define ARROW_STYLE_CLOSED             2
#define ARROW_STYLE_DOT                3
#define ARROW_STYLE_BOX                4
#define ARROW_STYLE_TICK               5

/* NOTE: Allow this enum to evaluate false */
#define LINE_STYLE_NO_LINE             0
#define LINE_STYLE_FLARED              1
#define LINE_STYLE_FLETCHING           2

#define RGB_BACKGROUND                 0
#define RGB_CROSSHAIR                  1
#define RGB_GRID                       2

#define ROTATE_NORMAL                  0
#define ROTATE_REFERENCE               1

#define SCALE_NORMAL                   0
#define SCALE_REFERENCE                1

#define HEART_NUM_POINTS               0
#define HEART_STYLE                    1
#define HEART_XSCALE                   2
#define HEART_YSCALE                   3

#define HEART_STYLE_4                  0
#define HEART_STYLE_5                  1

#define STAR_NUM_POINTS                0
#define STAR_CENTER_PT                 1
#define STAR_RAD_OUTER                 2
#define STAR_RAD_INNER                 3

#define DOLPHIN_NUM_POINTS             0
#define DOLPHIN_XSCALE                 1
#define DOLPHIN_YSCALE                 2

#define SNOWFLAKE_NUM_POINTS           0
#define SNOWFLAKE_XSCALE               1
#define SNOWFLAKE_YSCALE               2

/* Custom Data used in QGraphicsItems
*
*                     (     int, const QVariant)
* I.E. object.setData(OBJ_TYPE, OBJ_LINE);
* I.E. object.setData(OBJ_LAYER, "OUTLINE");
* I.E. object.setData(OBJ_COLOR, 123);
* I.E. object.setData(OBJ_LTYPE, OBJ_LTYPE_CONT);
*/

/* Keys */
/* value type - int:  See OBJ_VALUES */
// #define OBJ_TYPE                        0
/* value type - str:  See OBJ_NAME_VALUES */
//#define OBJ_NAME                       1
/* value type - str:  "USER", "DEFINED", "STRINGS" etc...  */
// #define OBJ_LAYER                       2
/* value type - int:  0-255 */
/* TODO: Use color chart in formats/format-dxf.h for this */
// #define OBJ_COLOR                       3
/* value type - int:  See OBJ_LTYPE_VALUES */
// #define OBJ_LTYPE                       4
/* value type - int:  0-27 */
// #define OBJ_LWT                         5
/* value type - int:  See RUBBER_VALUES */
// #define OBJ_RUBBER                      6

/* Values
* NOTE: Allow this enum to evaluate false
* NOTE: Values >= 65536 ensure compatibility with qgraphicsitem_cast()
```

```
*/
#define OBJ_NULL                        0
#define OBJ_BASE                   100000
#define OBJ_ARC                    100001
#define OBJ_BLOCK                  100002
#define OBJ_CIRCLE                 100003
#define OBJ_DIMALIGNED             100004
#define OBJ_DIMANGULAR             100005
#define OBJ_DIMARCLENGTH           100006
#define OBJ_DIMDIAMETER            100007
#define OBJ_DIMLEADER              100008
#define OBJ_DIMLINEAR              100009
#define OBJ_DIMORDINATE            100010
#define OBJ_DIMRADIUS              100011
#define OBJ_ELLIPSE                100012
#define OBJ_ELLIPSEARC             100013
#define OBJ_RUBBER                 100014
#define OBJ_GRID                   100015
#define OBJ_HATCH                  100016
#define OBJ_IMAGE                  100017
#define OBJ_INFINITELINE           100018
#define OBJ_LINE                   100019
#define OBJ_PATH                   100020
#define OBJ_POINT                  100021
#define OBJ_POLYGON                100022
#define OBJ_POLYLINE               100023
#define OBJ_RAY                    100024
#define OBJ_RECTANGLE              100025
#define OBJ_SLOT                   100026
#define OBJ_SPLINE                 100027
#define OBJ_TEXTMULTI              100028
#define OBJ_TEXTSINGLE             100029
#define OBJ_UNKNOWN                100030

/* OBJ_LTYPE_VALUES */
/* CAD Linetypes */
#define OBJ_LTYPE_CONT                  0
#define OBJ_LTYPE_CENTER                1
#define OBJ_LTYPE_DOT                   2
#define OBJ_LTYPE_HIDDEN                3
#define OBJ_LTYPE_PHANTOM               4
#define OBJ_LTYPE_ZIGZAG                5
/* Embroidery Stitchtypes */
/* __________ */
#define OBJ_LTYPE_RUNNING               6
/* vvvvvvvvvv */
#define OBJ_LTYPE_SATIN                 7
/* »»»»»» */
#define OBJ_LTYPE_FISHBONE              8

/* OBJ_LWT_VALUES */
#define OBJ_LWT_BYLAYER                -2
#define OBJ_LWT_BYBLOCK                -1
#define OBJ_LWT_DEFAULT                 0
#define OBJ_LWT_01                      1
#define OBJ_LWT_02                      2
#define OBJ_LWT_03                      3
#define OBJ_LWT_04                      4
#define OBJ_LWT_05                      5
#define OBJ_LWT_06                      6
#define OBJ_LWT_07                      7
#define OBJ_LWT_08                      8
#define OBJ_LWT_09                      9
#define OBJ_LWT_10                     10
#define OBJ_LWT_11                     11
#define OBJ_LWT_12                     12
#define OBJ_LWT_13                     13
#define OBJ_LWT_14                     14
#define OBJ_LWT_15                     15
#define OBJ_LWT_16                     16
#define OBJ_LWT_17                     17
#define OBJ_LWT_18                     18
#define OBJ_LWT_19                     19
#define OBJ_LWT_20                     20
#define OBJ_LWT_21                     21
#define OBJ_LWT_22                     22
#define OBJ_LWT_23                     23
#define OBJ_LWT_24                     24

/* OBJ_SNAP_VALUES NOTE: Allow this enum to evaluate false */
#define OBJ_SNAP_NULL                   0
#define OBJ_SNAP_ENDPOINT               1
#define OBJ_SNAP_MIDPOINT               2
#define OBJ_SNAP_CENTER                 3
#define OBJ_SNAP_NODE                   4
#define OBJ_SNAP_QUADRANT               5
#define OBJ_SNAP_INTERSECTION           6
```

```
#define OBJ_SNAP_EXTENSION             7
#define OBJ_SNAP_INSERTION             8
#define OBJ_SNAP_PERPENDICULAR         9
#define OBJ_SNAP_TANGENT              10
#define OBJ_SNAP_NEAREST              11
#define OBJ_SNAP_APPINTERSECTION      12
#define OBJ_SNAP_PARALLEL             13

/* NOTE: Allow this enum to evaluate true and false */
#define RUBBER_OFF                     0
#define RUBBER_ON                      1

#define RUBBER_CIRCLE_1P_RAD           2
#define RUBBER_CIRCLE_1P_DIA           3
#define RUBBER_CIRCLE_2P               4
#define RUBBER_CIRCLE_3P               5
#define RUBBER_CIRCLE_TTR              6
#define RUBBER_CIRCLE_TTT              7

#define RUBBER_DIMLEADER_LINE          8

#define RUBBER_ELLIPSE_LINE            9
#define RUBBER_ELLIPSE_MAJDIA_MINRAD  10
#define RUBBER_ELLIPSE_MAJRAD_MINRAD  11
#define RUBBER_ELLIPSE_ROTATION       12

#define RUBBER_GRIP                   13

#define RUBBER_LINE                   14

#define RUBBER_POLYGON                15
#define RUBBER_POLYGON_INSCRIBE       16
#define RUBBER_POLYGON_CIRCUMSCRIBE   17

#define RUBBER_POLYLINE               18

#define RUBBER_IMAGE                  19

#define RUBBER_RECTANGLE              20

#define RUBBER_TEXTSINGLE             21

/* Statusbar Buttons */
#define SB_SNAP                        0
#define SB_GRID                        1
#define SB_RULER                       2
#define SB_ORTHO                       3
#define SB_POLAR                       4
#define SB_QSNAP                       5
#define SB_QTRACK                      6
#define SB_LWT                         7
#define N_SB_BUTTONS                   8

/* NOTE: Allow this enum to evaluate false */
#define SPARE_RUBBER_OFF               0
#define SPARE_RUBBER_PATH              1
#define SPARE_RUBBER_POLYGON           2
#define SPARE_RUBBER_POLYLINE          3

/* NOTE: Allow this enum to evaluate false */
#define PREVIEW_CLONE_NULL             0
#define PREVIEW_CLONE_SELECTED         1
#define PREVIEW_CLONE_RUBBER           2

/* NOTE: Allow this enum to evaluate false */
#define PREVIEW_NULL              0
#define PREVIEW_MOVE              1
#define PREVIEW_ROTATE            2
#define PREVIEW_SCALE             3

/* Used for general int tables.  */
#define TERMINATOR_SYMBOL            -1

/* Menus:  matches the order of menu_list.  */
#define MENU_FILE                      0
#define MENU_EDIT                      1
#define MENU_VIEW                      2
#define MENU_DRAW                      3
#define MENU_TOOLS                     4
#define MENU_MODIFY                    5
#define MENU_SANDBOX                   6
#define MENU_DIMENSION                 7
#define MENU_WINDOW                    8
#define MENU_HELP                      9
#define MENU_RECENT                   10
#define MENU_ZOOM                     11
#define MENU_PAN                      12
```

```
#define N_MENUS                       13

/* Toolbars:  matches the order of toolbar_list.  */
#define TOOLBAR_FILE                   0
#define TOOLBAR_EDIT                   1
#define TOOLBAR_VIEW                   2
#define TOOLBAR_ICON                   3
#define TOOLBAR_HELP                   4
#define TOOLBAR_LAYER                  5
#define TOOLBAR_PROPERTIES             6
#define TOOLBAR_TEXT                   7
#define TOOLBAR_PROMPT                 8
#define TOOLBAR_DRAW                   9
#define TOOLBAR_MODIFY                10
#define TOOLBAR_INQUIRY               11
#define TOOLBAR_DIMENSION             12
#define TOOLBAR_SANDBOX               13
#define N_TOOLBARS                    14
#define TOOLBAR_BREAK                 -2

/* Settings */
#define GENERAL_LANGUAGE               0
#define GENERAL_ICON_THEME             1
#define GENERAL_ICON_SIZE              2
#define GENERAL_MDI_BG_USE_LOGO        3
#define GENERAL_MDI_BG_USE_TEXTURE     4
#define GENERAL_MDI_BG_USE_COLOR       5
#define GENERAL_MDI_BG_LOGO            6
#define GENERAL_MDI_BG_TEXTURE         7
#define GENERAL_MDI_BG_COLOR           8
#define GENERAL_TIP_OF_THE_DAY         9
#define GENERAL_CURRENT_TIP           10
#define GENERAL_SYSTEM_HELP_BROWSER   11
#define GENERAL_CHECK_FOR_UPDATES     12
#define DISPLAY_USE_OPENGL            13
#define DISPLAY_RENDERHINT_AA         14
#define DISPLAY_RENDERHINT_TEXT_AA    15
#define DISPLAY_RENDERHINT_SMOOTH_PIX 16
#define DISPLAY_RENDERHINT_HIGH_AA    17
#define DISPLAY_RENDERHINT_NONCOSMETIC 18
#define DISPLAY_SHOW_SCROLLBARS       19
#define DISPLAY_SCROLLBAR_WIDGET_NUM  20
#define DISPLAY_CROSSHAIR_COLOR       21
#define DISPLAY_BG_COLOR              22
#define DISPLAY_SELECTBOX_LEFT_COLOR  23
#define DISPLAY_SELECTBOX_LEFT_FILL   24
#define DISPLAY_SELECTBOX_RIGHT_COLOR 25
#define DISPLAY_SELECTBOX_RIGHT_FILL  26
#define DISPLAY_SELECTBOX_ALPHA       27
#define DISPLAY_ZOOMSCALE_IN          28
#define DISPLAY_ZOOMSCALE_OUT         29
#define DISPLAY_CROSSHAIR_PERCENT     30
#define DISPLAY_UNITS                 31
#define PROMPT_TEXT_COLOR             32
#define PROMPT_BG_COLOR               33
#define PROMPT_FONT_FAMILY            34
#define PROMPT_FONT_STYLE             35
#define PROMPT_FONT_SIZE              36
#define PROMPT_SAVE_HISTORY           37
#define PROMPT_SAVE_HISTORY_AS_HTML   38
#define PROMPT_SAVE_HISTORY_FILENAME  39
#define OPENSAVE_CUSTOM_FILTER        40
#define OPENSAVE_OPEN_FORMAT          41
#define OPENSAVE_OPEN_THUMBNAIL       42
#define OPENSAVE_SAVE_FORMAT          43
#define OPENSAVE_SAVE_THUMBNAIL       44
#define OPENSAVE_RECENT_MAX_FILES     45
#define OPENSAVE_RECENT_LIST_OF_FILES 46
#define OPENSAVE_RECENT_DIRECTORY     47
#define OPENSAVE_TRIM_DST_NUM_JUMPS   48
#define PRINTING_DEFAULT_DEVICE       49
#define PRINTING_USE_LAST_DEVICE      50
#define PRINTING_DISABLE_BG           51
#define GRID_SHOW_ON_LOAD             52
#define GRID_SHOW_ORIGIN              53
#define GRID_COLOR_MATCH_CROSSHAIR    54
#define GRID_COLOR                    55
#define GRID_LOAD_FROM_FILE           56
#define GRID_TYPE                     57
#define GRID_CENTER_ON_ORIGIN         58
#define GRID_CENTER_X                 59
#define GRID_CENTER_Y                 60
#define GRID_SIZE_X                   61
#define GRID_SIZE_Y                   62
#define GRID_SPACING_X                63
#define GRID_SPACING_Y                64
#define GRID_SIZE_RADIUS              65
```

```
#define GRID_SPACING_RADIUS           66
#define GRID_SPACING_ANGLE            67
#define RULER_SHOW_ON_LOAD            68
#define RULER_METRIC                  69
#define RULER_COLOR                   70
#define RULER_PIXEL_SIZE              71
#define QSNAP_ENABLED                 72
#define QSNAP_LOCATOR_COLOR           73
#define QSNAP_LOCATOR_SIZE            74
#define QSNAP_APERTURE_SIZE           75
#define QSNAP_ENDPOINT                76
#define QSNAP_MIDPOINT                77
#define QSNAP_CENTER                  78
#define QSNAP_NODE                    79
#define QSNAP_QUADRANT                80
#define QSNAP_INTERSECTION            81
#define QSNAP_EXTENSION               82
#define QSNAP_INSERTION               83
#define QSNAP_PERPENDICULAR           84
#define QSNAP_TANGENT                 85
#define QSNAP_NEAREST                 86
#define QSNAP_APPARENT                87
#define QSNAP_PARALLEL                88
#define LWT_SHOW_LWT                  89
#define LWT_REAL_RENDER               90
#define LWT_DEFAULT_LWT               91
#define SELECTION_MODE_PICKFIRST      92
#define SELECTION_MODE_PICKADD        93
#define SELECTION_MODE_PICKDRAG       94
#define SELECTION_COOLGRIP_COLOR      95
#define SELECTION_HOTGRIP_COLOR       96
#define SELECTION_GRIP_SIZE           97
#define SELECTION_PICKBOX_SIZE        98
#define TEXT_FONT                     99
#define TEXT_SIZE                    100
#define TEXT_ANGLE                   101
#define TEXT_STYLE_BOLD              102
#define TEXT_STYLE_ITALIC            103
#define TEXT_STYLE_UNDERLINE         104
#define TEXT_STYLE_OVERLINE          105
#define TEXT_STYLE_STRIKEOUT         106
#define N_SETTINGS                   107

/* Settings memory */
#define SETTING_SETTING                0
#define SETTING_DIALOG                 1
#define SETTING_ACCEPT                 2
#define SETTING_PREVIEW                3

/* Justification */
#define JUSTIFY_LEFT                   0
#define JUSTIFY_CENTER                 1
#define JUSTIFY_RIGHT                  2
#define JUSTIFY_ALIGNED                3
#define JUSTIFY_MIDDLE                 4
#define JUSTIFY_FIT                    5
#define JUSTIFY_TOP_LEFT               6
#define JUSTIFY_TOP_CENTER             7
#define JUSTIFY_TOP_RIGHT              8
#define JUSTIFY_MIDDLE_LEFT            9
#define JUSTIFY_MIDDLE_CENTER         10
#define JUSTIFY_MIDDLE_RIGHT          11
#define JUSTIFY_BOTTOM_LEFT           12
#define JUSTIFY_BOTTOM_CENTER         13
#define JUSTIFY_BOTTOM_RIGHT          14
#define N_JUSTIFY                     15

/* Object Properties */
#define PROP_ARC_CENTER_X              0
#define PROP_ARC_CENTER_Y              1
#define PROP_ARC_RADIUS                2
#define N_PROPS                        3

/* View Properties */
#define VIEW_SNAP                      0
#define VIEW_GRID                      1
#define VIEW_ORTHO                     2
#define VIEW_POLAR                     3
#define VIEW_QSNAP                     4
#define VIEW_QTRACK                    5
#define VIEW_LWT                       6
#define VIEW_REAL                      7
#define VIEW_RULER                     8
#define VIEW_METRIC                    9
#define VIEW_GRIPPING                 10
#define VIEW_RAPID_MOVING             11
#define VIEW_PREVIEWING               12
```

```
#define VIEW_PASTING                  13
#define VIEW_MOVING                   14
#define VIEW_SELECTING                15
#define VIEW_ZOOMING                  16
#define VIEW_PANNING_RT               17
#define VIEW_PANNING_POINT            18
#define VIEW_PANNING                  19
#define VIEW_GRID_COLOR               20
#define VIEW_RULER_COLOR              21
#define VIEW_CROSSHAIR_COLOR          22
#define N_VIEW_ATTR                   23

/* Translations */
#define UNFINISHED                     0
#define DRAFT                          1
#define FINISHED                       2

#define ARABIC_SHORTCODE            "ar"
#define CHINESE_SHORTCODE           "zh"
#define ENGLISH_SHORTCODE           "en"
#define FRENCH_SHORTCODE            "fr"
#define RUSSIAN_SHORTCODE           "ru"
#define SPANISH_SHORTCODE           "es"

#define GB_GENERAL                     0
#define GB_GEOM_ARC                    1
#define GB_MISC_ARC                    2
#define GB_GEOM_BLOCK                  3
#define GB_GEOM_CIRCLE                 4
#define GB_GEOM_DIMALIGNED             5
#define GB_GEOM_DIMANGULAR             6
#define GB_GEOM_DIMARCLENGTH           7
#define GB_GEOM_DIMDIAMETER            8
#define GB_GEOM_DIMLEADER              9
#define GB_GEOM_DIMLINEAR             10
#define GB_GEOM_DIMORDINATE           11
#define GB_GEOM_DIMRADIUS             12
#define GB_GEOM_ELLIPSE               13
#define GB_GEOM_IMAGE                 14
#define GB_MISC_IMAGE                 15
#define GB_GEOM_INFINITE_LINE         16
#define GB_GEOM_LINE                  17
#define GB_GEOM_PATH                  18
#define GB_MISC_PATH                  19
#define GB_GEOM_POINT                 20
#define GB_GEOM_POLYGON               21
#define GB_GEOM_POLYLINE              22
#define GB_MISC_POLYLINE              23
#define GB_GEOM_RAY                   24
#define GB_GEOM_RECT                  25
#define GB_GEOM_TEXT_MULTI            26
#define GB_TEXT_TEXT_SINGLE           27
#define GB_GEOM_TEXT_SINGLE           28
#define GB_MISC_TEXT_SINGLE           29
#define N_GROUPBOX                    30

#define REAL(arg) context->argument[arg].r
#define INT(arg)  context->argument[arg].i
#define STR(arg)  context->argument[arg].s
#define BOOL(arg) context->argument[arg].b

#define DIALOG_INT_SLOT(A) \
SLOT([=](int value) { setting[A].dialog.i = value; } )
#define DIALOG_REAL_SLOT(A) \
SLOT([=](EmbReal value) { setting[A].dialog.r = value; } )
#define DIALOG_STRING_SLOT(A) \
SLOT([=](QString value) { strncpy(setting[A].dialog.s, qPrintable(value), MAX_STRING_LENGTH); } )

/* For switch tables we can use this trick to use two character indices.
* For example:  "case TWO_CHAR_INDEX(DUTCH_SHORTCODE):".
*/
#define TWO_CHAR_INDEX(A)             (0x100*A[0] + A[1])

typedef ScriptValue (*Command)(ScriptEnv *context);

typedef struct CommandData_ {
    char *command;
    char *arguments;
    char *icon;
    char *tooltip;
    char *statustip;
    char *alias;
    char *shortcut;
    int32_t flags;
    Command action;
} CommandData;

```

```
#define MAX_LAYERS 20

typedef struct LayerData_ {
    char name[200];
    bool visible;
    bool frozen;
    EmbReal zValue;
    int32_t color;
    char *lineType;
    char *lineWeight;
    bool print;
} LayerData;

typedef struct Setting_ {
    char key[MAX_STRING_LENGTH];
    ScriptValue setting;
    ScriptValue dialog;
    ScriptValue accept;
    ScriptValue preview;
} Setting;

typedef struct GroupBoxData_ {
    int32_t id;
    char *key;
    char *label;
    int rows;
    char *data[100];
    int object;
} GroupBoxData;

typedef struct ObjectCore_ {
    int32_t doc;

    EmbGeometry *geometry;

    LabelledVector *rubber_points;
    int n_rubber_points;
    StringMap *rubber_texts;
    int n_rubber_texts;

    EmbVector position;

    EmbReal scale;
    EmbReal rotation;

    uint32_t rgb;

    char OBJ_NAME[MAX_STRING_LENGTH];

    char text[MAX_STRING_LENGTH];
    char textFont[MAX_STRING_LENGTH];
    char textJustify[MAX_STRING_LENGTH];
    EmbReal textSize;
    bool textBold;
    bool textItalic;
    bool textUnderline;
    bool textStrikeOut;
    bool textOverline;
    bool textBackward;
    bool textUpsideDown;

    int rubber_mode;
    uint32_t objID;

    bool curved;
    bool filled;
    EmbReal arrowStyleAngle;
    EmbReal arrowStyleLength;
    EmbReal lineStyleAngle;
    EmbReal lineStyleLength;

    int gripIndex;
} ObjectCore;

typedef struct UndoData_ {
    char label[MAX_STRING_LENGTH];
    int type;
    int32_t doc;
    EmbVector delta;
    EmbVector pivot;
    EmbReal angle;
    EmbReal factor;
    char navType[MAX_STRING_LENGTH];
    EmbVector fromCenter;
    EmbVector toCenter;
    bool done;
    EmbVector before;
```

```
    EmbVector after;
    EmbLine mirrorLine;
    uint32_t obj;
} UndoData;

/* .  */
typedef struct DocumentData_ {
    int32_t id;
    EmbPattern *pattern;
    ScriptValue attributes[N_VIEW_ATTR];
    UndoData undo_stack[MAX_UNDO];

    EmbIdList *selectedItems;
    EmbIdList *rubberRoomList;
    EmbIdList *previewObjectList;
    EmbIdList *spareRubberList;
    EmbIdList *cutCopyObjectList;
    EmbIdList *previewObjectItemGroup;
    EmbIdList *hashDeletedObjects;
    EmbIdList *pasteObjectItemGroup;

    EmbVectorList *selectedGripPoints;

    /* UI State */
    bool qSnapActive;
    bool qSnapToggle;

    EmbVector previewPoint;
    EmbReal previewData;
    int previewMode;

    EmbVector viewMousePoint;
    EmbVector sceneMousePoint;
    EmbVector sceneQSnapPoint;

    uint8_t qsnapLocatorSize;
    uint32_t qsnapLocatorColor;
    uint8_t qsnapApertureSize;

    uint32_t gripColorCool;
    uint32_t gripColorHot;
    uint32_t gridColor;
    uint8_t gripSize;
    uint8_t pickBoxSize;

    uint32_t crosshairColor;
    uint32_t crosshairSize;

    EmbVector scenePressPoint;
    EmbVector pressPoint;
    EmbVector sceneMovePoint;
    EmbVector movePoint;
    EmbVector sceneReleasePoint;
    EmbVector releasePoint;
    EmbVector sceneGripPoint;

    EmbVector cutCopyMousePoint;
    EmbVector pasteDelta;

    int panDistance;
    int panStartX;
    int panStartY;

    uint32_t rulerColor;
    bool rulerMetric;
    uint8_t rulerPixelSize;

    uint32_t backgroundColor;

    bool fileWasLoaded;
    int myIndex;

    char curFile[MAX_STRING_LENGTH];
    char curLayer[MAX_STRING_LENGTH];
    uint32_t curColor;
    char curLineType[MAX_STRING_LENGTH];
    char curLineWeight[MAX_STRING_LENGTH];
} DocumentData;

/* For dialogs and tabs */
typedef struct WidgetDesc_ {
    int id;
    char *label;
    char *key;
    char *icon;
    char *value;
     char *editor_data;
```

```
    int type;
    int n_leaves;
} WidgetDesc;

typedef struct Button_ {
    EmbRect rect;
    EmbColor color;
    EmbColor highlight_color;
    char *text;
    char *font;
    EmbColor text_color;
    int state;
} Button;

typedef struct Tab_ {
    int state;
} Tab;

typedef struct Translation_ {
    char *ar;
    char *en;
    char *es;
    char *fr;
    char *ru;
    char *zh;
    int32_t status;
} Translation;

typedef struct Widget_ {
    EmbString key;
    void *widget;
} Widget;

typedef struct WidgetData_ {
    char *key;
    char *label;
    int id;
    int type;
} WidgetData;

typedef struct ViewData_ {
    int32_t doc;
    double little;
    double medium;
    EmbLine *lines;
    int n_lines;
    EmbGeometry *geometry;
    int n_geometry;
    int width;
    int height;
    int unit;
    bool feet;
    double fraction;
    int xStart;
    int yStart;
} ViewData;

/* ------------------------------- Scripting ----------------------------
* This is gradually growing to incorporate our full data structure so
* no global data is stored as externs.
*/
ScriptEnv *create_script_env(void);
void free_script_env(ScriptEnv *);
ScriptEnv *pack(ScriptEnv *context, const char *fmt, ...);
ScriptValue call(ScriptEnv *context, char *function, ...);
ScriptValue run_cmd(ScriptEnv *context, const char *line);

ScriptValue command_prompt(ScriptEnv *context, const char *line);

void set_bool(int32_t key, bool value);
void set_int(int32_t key, int32_t value);
void set_real(int32_t key, EmbReal value);
void set_str(int32_t key, const char *value);

bool get_bool(int32_t key);
int32_t get_int(int32_t key);
EmbReal get_real(int32_t key);
const char *get_str(int32_t key);

EmbVector find_mouse_snap_point(EmbVectorList *snap_points, EmbVector mouse_point);

ViewData *create_view_data(int32_t doc, int width, int height, int unit);
void free_view_data(ViewData *vdata);
int create_ruler_lines(ViewData *data);

bool string_equal(char *a, const char *b);
void string_copy(char *dst, const char *src);
```

```
int find_in_map(StringMap *hash, int length, const char *key);
int table_length(char *table[]);

EmbVectorList *create_vector_list(void);
void append_vector_to_list(EmbVectorList *, EmbVector);
void remove_vector_from_list(EmbVectorList *, int32_t);
void free_vector_list(EmbVectorList *);

EmbIdList *create_id_list(void);
void append_id_to_list(EmbIdList *list, int32_t);
void remove_id_from_list(EmbIdList *list, int32_t);
bool id_list_contains(EmbIdList *list, int32_t);
void copy_object_list(EmbIdList *dst, EmbIdList *src);
void free_id_list(EmbIdList *);

ScriptValue *setting_ptr(int key, int mode);
void copy_setting(int key, int dst, int src);

void prompt_output(const char *);
int argument_checks(ScriptEnv *context, int id);
char *translate(const char *msg);

bool pattern_save(EmbPattern *pattern, char *fileName);

int parse_floats(const char *line, float result[], int n);
int parse_vector(const char *line, EmbVector *v);
bool valid_rgb(float r, float g, float b);
void report_distance(EmbVector a, EmbVector b);

void add_to_menu(int id, char *menu_data[]);
void add_to_toolbar(int id, char *toolbar_data[]);

int load_data(void);

int load_settings(const char *appDir, const char *configDir);
int save_settings(char *appDir, char *configDir);

int get_command_id(char *);

EmbArc emb_arc_set_radius(EmbArc a, EmbReal radius);

bool int32_underflow(int64_t a, int64_t b);
bool int32_overflow(int64_t a, int64_t b);
int round_to_multiple(bool roundUp, int numToRound, int multiple);

void messagebox(const char* logo, const char *title, const char *text);

const char *get_current_layer(void);
uint32_t get_current_color(void);
const char *get_current_line_type(void);
const char *get_current_line_weight(void);

void statusbar_toggle(const char *key, bool on);
void zoom_extents_all_sub_windows(void);
bool loadFile(const char *fileName);

int32_t free_glfw(void);

int glfw_application(int argc, char *argv[]);

int find_int_map(IntMap *, int);

char *file_extension(char *fileName);

/* ----------------------------- Widgets ------------------------------ */

void set_visibility(const char *key, bool visibility);
void set_enabled(const char *key, bool visibility);
void set_visibility_group(char *key[], bool visibility);
void set_enabled_group(char *key[], bool visibility);

/* ----------------------------- Prompt ------------------------------ */

void set_prompt_text_color(uint32_t color);
void set_prompt_background_color(uint32_t color);
void set_prompt_font_family(char *family);
void set_prompt_font_style(char *style);
void set_prompt_font_size(int size);

void prompt_history_appended(char *txt);
void log_prompt_input(char *txt);
void prompt_input_previous(void);
void prompt_input_next(void);
void prompt_update_style(void);
void prompt_end_command(void);
void prompt_set_current_text(const char *);
void process_input(char);
```

```

/* ------------------------- Main Functions -------------------------- */

void run_command_main(const char *cmd);
void run_command_click(const char *cmd, EmbReal x, EmbReal y);
void run_command_move(const char *cmd, EmbReal x, EmbReal y);
void run_command_context(const char *cmd, const char *str);
void run_command_prompt(const char *cmd);

uint32_t rgb(uint8_t r, uint8_t g, uint8_t b);

void update_all_view_scroll_bars(bool val);
void update_all_view_cross_hair_colors(uint32_t color);
void update_all_view_background_colors(uint32_t color);
void update_all_view_select_box_colors(uint32_t colorL, uint32_t fillL,
    uint32_t colorR, uint32_t fillR, int alpha);
void update_all_view_grid_colors(uint32_t color);
void update_all_view_ruler_colors(uint32_t color);

void update_pick_add_mode(bool val);
void pick_add_mode_toggled(void);

void set_text_font(char *str);
void set_text_size(EmbReal num);

void prompt_history_appended(char *txt);
void log_prompt_input(char *txt);

void open_file(bool recent, char *recentFile);
void open_selected_files(char *table[]);

void settings_dialog(const char *showTab);

bool valid_file_format(char *fileName);

void set_undo_clean_icon(bool opened);

void mdiarea_set_bg(uint32_t color);
void accept_interface_color(int32_t key, uint32_t color);
void preview_interface_color(int32_t key, uint32_t color);
void dialog_interface_color(int32_t key, uint32_t color);

void current_layer_changed(char *layer);
void current_linetype_changed(char *type);
void current_lineweight_changed(char *weight);

void prompt_input_previous(void);
void prompt_input_next(void);

void layer_selector_changed(int index);
void linetype_selector_changed(int index);
void lineweight_selector_changed(int index);
void text_size_selector_index_changed(int index);

void make_layer_active(void);
void layer_manager(void);
void layer_previous(void);

void delete_pressed(void);
void escape_pressed(void);

void icon_resize(int iconSize);

void read_settings(void);
void write_settings(void);

void create_all_actions(void);

void load_formats(void);

void settings_prompt(void);

void end_command(void);
void wait_cursor(void);
void arrow_cursor(void);
void restore_cursor(void);

void cut(void);

void about_dialog(void);

/* ------------------------------- Editors ------------------------------- */

void check_box_lwt_real_render_changed(int);
void combo_box_prompt_font_family_changed(char *);
void combo_box_prompt_font_style_changed(char *);
void spin_box_display_select_box_alpha_changed(int);
```

```
void spin_box_prompt_font_size_changed(int);

void checkBoxGridCenterOnOriginStateChanged(int);
void comboBoxGridTypeCurrentIndexChanged(const char *);
void checkBoxGridColorMatchCrossHairStateChanged(int);
void checkBoxGridLoadFromFileStateChanged(int);
void checkBoxLwtShowLwtStateChanged(int);

void accept_settings(void);

void useBackgroundLogo(bool use);
void useBackgroundTexture(bool use);
void useBackgroundColor(bool use);

void setBackgroundLogo(const char *fileName);
void setBackgroundTexture(const char *fileName);
void setBackgroundColor(uint32_t color);

void update_view(void);
void preview_update(void);

/* ------------------------------- Commands ------------------------------- */

void stub_testing(void);
void run_testing(void);
void check_for_updates(void);
void open_recent_file(void);
int save_as_file(void);
void update_interface(void);
void window_menu_about_to_show(void);
void start_blinking(void);
void stop_blinking(void);
void repeat_action(void);
void move_action(void);

void nanosleep_(int);

void button_tip_of_the_day_clicked(int button);

void set_mouse_coord(EmbReal x, EmbReal y);

void set_background_color(uint8_t r, uint8_t g, uint8_t b);
void set_cross_hair_color(uint8_t r, uint8_t g, uint8_t b);
void set_grid_color(uint8_t r, uint8_t g, uint8_t b);

void set_cursor_shape(char *shape);

void create_details_dialog(void);

void set_prompt_prefix(char *txt);
/* TODO: void set_RubberFilter(int64_t id); */
/* TODO: This is so more than 1 rubber object can exist at one time without updating all rubber objects at once.  */
void set_rubber_mode(int mode);
void set_rubber_point(const char *key, EmbReal x, EmbReal y);
void set_rubber_text(const char *key, char *txt);

void toggle(int mode);

/* Help Menu */
void button_tip_of_the_day_clicked(int);
void check_box_tip_of_the_day_changed(int);
void whats_this_context_help(void);

int make_application(int argc, char* argv[]);

void set_object_rubber_mode(ObjectCore *core, int mode);

EmbVector unpack_vector(ScriptEnv *context, int offset);

void undoable_add(int32_t doc, uint32_t obj, char *label);
void undoable_delete(int32_t doc, uint32_t obj, char *label);
void undoable_scale(int32_t doc, uint32_t obj, EmbVector center, EmbReal factor, char *label);
void undoable_move(int32_t doc, uint32_t obj, EmbVector delta, char *msg);
void undoable_rotate(int32_t doc, uint32_t obj, EmbVector v, char *msg);
void undoable_mirror(int32_t doc, uint32_t obj, EmbVector start, EmbVector end,
    char *msg);

void update_all_background_color(uint32_t color);

void create_all_menus(void);
void create_all_toolbars(void);

void clear_all_fields(void);
void line_edit_clear(const char *key);
void combo_box_clear(const char *key);
void clear_font_combobox(void);
```

```
int get_id(char *table[], char *);

const char *platform_string(void);

void button_tip_of_the_day_clicked(int button);
void free_objects(EmbIdList *);

/* All of the pointers used by command_data.
*
* Note that these are not global scope functions because they are only needed
* for this table.
*/
ScriptValue about_command(ScriptEnv *context);
ScriptValue add_arc_command(ScriptEnv *context);
ScriptValue add_circle_command(ScriptEnv *context);
ScriptValue add_dimleader_command(ScriptEnv *context);
ScriptValue add_ellipse_command(ScriptEnv *context);
ScriptValue add_horizontal_dimension_command(ScriptEnv *context);
ScriptValue add_image_command(ScriptEnv *context);
ScriptValue add_infinite_line_command(ScriptEnv *context);
ScriptValue add_line_command(ScriptEnv *context);
ScriptValue add_ray_command(ScriptEnv *context);
ScriptValue add_triangle_command(ScriptEnv *context);
ScriptValue add_rectangle_command(ScriptEnv *context);
ScriptValue add_rounded_rectangle_command(ScriptEnv *context);
ScriptValue add_slot_command(ScriptEnv *context);
ScriptValue add_point_command(ScriptEnv *context);
ScriptValue add_regular_polygon_command(ScriptEnv *context);
ScriptValue add_vertical_dimension_command(ScriptEnv *context);
ScriptValue add_textmulti_command(ScriptEnv *context);
ScriptValue add_textsingle_command(ScriptEnv *context);
ScriptValue alert_command(ScriptEnv *context);
ScriptValue angle_command(ScriptEnv *context);
ScriptValue changelog_command(ScriptEnv *context);
ScriptValue copy_command(ScriptEnv *context);
ScriptValue color_selector_command(ScriptEnv *context);
ScriptValue current_color_changed_command(ScriptEnv *context);
ScriptValue debug_command(ScriptEnv *context);
ScriptValue details_command(ScriptEnv *context);
ScriptValue disable_command(ScriptEnv *context);
ScriptValue do_nothing_command(ScriptEnv *context);
ScriptValue enable_command(ScriptEnv *context);
ScriptValue exit_command(ScriptEnv *context);
ScriptValue get_command(ScriptEnv *context);
ScriptValue help_command(ScriptEnv *context);
ScriptValue layer_previous_command(ScriptEnv *context);
ScriptValue make_layer_active_command(ScriptEnv *context);
ScriptValue move_command(ScriptEnv *context);
ScriptValue new_command(ScriptEnv *context);
ScriptValue open_command(ScriptEnv *context);
ScriptValue paste_command(ScriptEnv *context);
ScriptValue previewon_command(ScriptEnv *context);
ScriptValue print_command(ScriptEnv *context);
ScriptValue redo_command(ScriptEnv *context);
ScriptValue sandbox_command(ScriptEnv *context);
ScriptValue save_command(ScriptEnv *context);
ScriptValue save_as_command(ScriptEnv *context);
ScriptValue set_command(ScriptEnv *context);
ScriptValue tip_of_the_day_command(ScriptEnv *context);
ScriptValue undo_command(ScriptEnv *context);
ScriptValue window_cascade_command(ScriptEnv *context);
ScriptValue window_close_command(ScriptEnv *context);
ScriptValue window_close_all_command(ScriptEnv *context);
ScriptValue window_next_command(ScriptEnv *context);
ScriptValue window_previous_command(ScriptEnv *context);
ScriptValue window_tile_command(ScriptEnv *context);
ScriptValue cut_command(ScriptEnv *context);
ScriptValue copy_selected_command(ScriptEnv *context);
ScriptValue paste_selected_command(ScriptEnv *context);
ScriptValue whats_this_command(ScriptEnv *context);
ScriptValue icon16_command(ScriptEnv *context);
ScriptValue icon24_command(ScriptEnv *context);
ScriptValue icon32_command(ScriptEnv *context);
ScriptValue icon48_command(ScriptEnv *context);
ScriptValue icon64_command(ScriptEnv *context);
ScriptValue icon128_command(ScriptEnv *context);
ScriptValue settings_command(ScriptEnv *context);
ScriptValue layers_command(ScriptEnv *context);
ScriptValue layer_selector_command(ScriptEnv *context);
ScriptValue line_type_selector_command(ScriptEnv *context);
ScriptValue line_weight_selector_command(ScriptEnv *context);
ScriptValue hide_all_layers_command(ScriptEnv *context);
ScriptValue show_all_layers_command(ScriptEnv *context);
ScriptValue freeze_all_layers_command(ScriptEnv *context);
ScriptValue thaw_all_layers_command(ScriptEnv *context);
ScriptValue lock_all_layers_command(ScriptEnv *context);
```

```
ScriptValue unlock_all_layers_command(ScriptEnv *context);
ScriptValue text_bold_command(ScriptEnv *context);
ScriptValue text_italic_command(ScriptEnv *context);
ScriptValue text_underline_command(ScriptEnv *context);
ScriptValue text_strikeout_command(ScriptEnv *context);
ScriptValue text_overline_command(ScriptEnv *context);
ScriptValue zoom_real_time_command(ScriptEnv *context);
ScriptValue zoom_previous_command(ScriptEnv *context);
ScriptValue zoom_window_command(ScriptEnv *context);
ScriptValue zoom_dynamic_command(ScriptEnv *context);
ScriptValue zoom_scale_command(ScriptEnv *context);
ScriptValue zoom_center_command(ScriptEnv *context);
ScriptValue zoom_in_command(ScriptEnv *context);
ScriptValue zoom_out_command(ScriptEnv *context);
ScriptValue zoom_selected_command(ScriptEnv *context);
ScriptValue zoom_all_command(ScriptEnv *context);
ScriptValue zoom_extents_command(ScriptEnv *context);
ScriptValue panrealtime_command(ScriptEnv *context);
ScriptValue panpoint_command(ScriptEnv *context);
ScriptValue panleft_command(ScriptEnv *context);
ScriptValue panright_command(ScriptEnv *context);
ScriptValue panup_command(ScriptEnv *context);
ScriptValue pandown_command(ScriptEnv *context);
ScriptValue day_command(ScriptEnv *context);
ScriptValue night_command(ScriptEnv *context);
ScriptValue circle_command(ScriptEnv *context);
ScriptValue dimleader_command(ScriptEnv *context);
ScriptValue distance_command(ScriptEnv *context);
ScriptValue dolphin_command(ScriptEnv *context);
ScriptValue ellipse_command(ScriptEnv *context);
ScriptValue erase_command(ScriptEnv *context);
ScriptValue error_command(ScriptEnv *context);
ScriptValue heart_command(ScriptEnv *context);
ScriptValue line_command(ScriptEnv *context);
ScriptValue located_point_command(ScriptEnv *context);
ScriptValue mirror_selected_command(ScriptEnv *context);
ScriptValue move_selected_command(ScriptEnv *context);
ScriptValue path_command(ScriptEnv *context);
ScriptValue platform_command(ScriptEnv *context);
ScriptValue point_command(ScriptEnv *context);
ScriptValue polygon_command(ScriptEnv *context);
ScriptValue polyline_command(ScriptEnv *context);
ScriptValue previewoff_command(ScriptEnv *context);
ScriptValue quickleader_command(ScriptEnv *context);
ScriptValue rectangle_command(ScriptEnv *context);
ScriptValue rgb_command(ScriptEnv *context);
ScriptValue rotate_command(ScriptEnv *context);
ScriptValue scale_command(ScriptEnv *context);
ScriptValue scale_selected_command(ScriptEnv *context);
ScriptValue select_all_command(ScriptEnv *context);
ScriptValue single_line_text_command(ScriptEnv *context);
ScriptValue snowflake_command(ScriptEnv *context);
ScriptValue star_command(ScriptEnv *context);
ScriptValue syswindows_command(ScriptEnv *context);
ScriptValue todo_command(ScriptEnv *context);
ScriptValue vulcanize_command(ScriptEnv *context);
ScriptValue add_geometry_command(ScriptEnv *context);
ScriptValue delete_command(ScriptEnv *context);
ScriptValue grip_edit_command(ScriptEnv *context);
ScriptValue nav_command(ScriptEnv *context);
ScriptValue mirror_command(ScriptEnv *context);
ScriptValue test_command(ScriptEnv *context);
ScriptValue simulate_command(ScriptEnv *context);
ScriptValue play_command(ScriptEnv *context);
ScriptValue stop_command(ScriptEnv *context);
ScriptValue pause_command(ScriptEnv *context);
ScriptValue fast_forward_command(ScriptEnv *context);
ScriptValue export_command(ScriptEnv *context);
ScriptValue qr_command(ScriptEnv *context);
ScriptValue lettering_command(ScriptEnv *context);
ScriptValue pattern_command(ScriptEnv *context);
ScriptValue design_command(ScriptEnv *context);


/* ------------------------------ Document ------------------------------- */

void doc_init(int32_t doc);
void free_doc(int32_t doc);

bool doc_allow_zoom_in(int32_t doc);
bool doc_allow_zoom_out(int32_t doc);
void doc_zoom_window(int32_t doc);
void doc_zoom_selected(int32_t doc);
void doc_zoom_to_point(int32_t doc, EmbVector mousePoint, int zoomDir);

void doc_nav(char *label, int32_t doc);
void doc_toggle(int32_t doc, const char *key, bool on);
```

```
void doc_select_all(int32_t doc);
void doc_selection_changed(int32_t doc);
void doc_clear_selection(int32_t doc);
void doc_delete_selected(int32_t doc);
void doc_move_selected(int32_t doc, EmbVector delta);
void doc_cut(int32_t doc);
void doc_copy(int32_t doc);
void doc_paste(int32_t doc);
void doc_repeat_action(int32_t doc);
void doc_move_action(int32_t doc);
void doc_scale_action(int32_t doc);
void doc_scale_selected(int32_t doc, EmbReal x, EmbReal y, EmbReal factor);
void doc_rotate_action(int32_t doc);
void doc_rotate_selected(int32_t doc, EmbVector position, EmbReal rot);
void doc_mirror_selected(int32_t doc, EmbReal x1, EmbReal y1, EmbReal x2, EmbReal y2);
int doc_num_selected(int32_t doc);

void doc_add_to_rubber_room(int32_t doc, int32_t item);

void doc_stop_gripping(int32_t, bool);
void hide_selectbox(int32_t);

ScriptValue *doc_get_attr(int32_t doc, uint32_t key);
bool doc_get_bool(int32_t doc, uint32_t key);
int32_t doc_get_int(int32_t doc, uint32_t key);
void doc_set_attr(int32_t doc, uint32_t key, ScriptValue value);
void doc_set_bool(int32_t doc, uint32_t key, bool value);
void doc_set_int(int32_t doc, uint32_t key, int32_t value);

void remove_paste_object_item_group(int32_t doc);
void doc_empty_grid(int32_t doc);
void doc_set_grid_color(int32_t doc, uint32_t color);
void doc_set_ruler_color(int32_t doc, uint32_t color);

void set_prompt_text_color(uint32_t color);
void set_prompt_background_color(uint32_t color);

void doc_preview_on(int32_t doc, int clone, int mode, EmbReal x, EmbReal y, EmbReal data);
void doc_preview_off(int32_t doc);

void doc_delete_pressed(int32_t doc);
void doc_escape_pressed(int32_t doc);

void doc_corner_button_clicked(int32_t doc);
void doc_create_grid(int32_t doc, const char *gridType);
void doc_copy_selected(int32_t doc);

void doc_show_scroll_bars(int32_t doc, bool val);
void doc_set_corner_button(int32_t doc);
void doc_set_cross_hair_color(int32_t doc, uint32_t color);
void doc_set_cross_hair_size(int32_t doc, uint8_t percent);
void doc_set_background_color(int32_t doc, uint32_t color);
void doc_set_select_box_colors(int32_t doc, uint32_t colorL, uint32_t fillL,
    uint32_t colorR, uint32_t fillR, int alpha);

bool doc_allow_rubber(int32_t doc);
void doc_vulcanize_rubber_room(int32_t doc);
void doc_clear_rubber_room(int32_t doc);
void doc_spare_rubber(int32_t doc, int64_t id);
void doc_set_rubber_mode(int32_t doc, int mode);
void doc_set_rubber_point(int32_t doc, char *key, EmbVector point);
void doc_set_rubber_text(int32_t doc, char *key, char *txt);

void doc_recalculate_limits(int32_t doc);
void doc_center_at(int32_t doc, EmbVector centerPoint);
EmbVector doc_center(int32_t doc);

void doc_create_grid_rect(int32_t doc);
void doc_create_grid_polar(int32_t doc);
void doc_create_grid_iso(int32_t doc);
void doc_create_origin(int32_t doc);

void doc_load_ruler_settings(int32_t doc);

void doc_update_mouse_coords(int32_t doc, int x, int y);

void doc_pan_start(int32_t doc, EmbVector point);

void doc_align_scene_point_with_view_point(int32_t doc, EmbVector scenePoint, EmbVector viewPoint);

void doc_undoable_add_obj(int32_t doc, uint32_t id, int rubberMode);

double doc_width(int32_t doc);
double doc_height(int32_t doc);

void doc_corner_button_clicked(int32_t doc);
```

```
void doc_add_item(int32_t doc, uint32_t id);
void doc_update(int32_t doc_index);

void doc_center_on(int32_t doc, EmbVector v);

DocumentData *doc_data(int32_t doc);
EmbVector doc_center(int32_t doc_id);
void doc_scale(int32_t doc_id, EmbReal s);
void doc_begin_macro(int32_t doc, char *s);
void doc_end_macro(int32_t doc);

void update_color_linetype_lineweight(void);

void rotate_action(void);
void scale_action(void);

EmbVector doc_map_from_scene(int32_t doc, EmbVector v);
EmbVector doc_map_to_scene(int32_t doc, EmbVector v);

void doc_add_object(int32_t doc, uint32_t obj);
void doc_delete_object(int32_t doc, uint32_t obj);
void doc_vulcanize_object(int32_t doc, uint32_t obj);

int32_t active_document(void);

void whats_this_mode(void);

void enable_rapid_fire(void);
void disable_rapid_fire(void);

/* ------------------------------- Interface ------------------------------ */

void create_properties_group_box(int32_t);

void add_command(char *alias, char *cmd);

void preview_update(void);
void update_editors(int32_t id);
void edit_field(int32_t id, const char *objName, const char *text);

void show_widget(const char *key, int type);
void hide_widget(const char *key, int type);
void widget_clear(const char *key, int type);
int find_widget_map(const char *key);

void show_groups(int objType);
void show_one_type(int index);
void hide_all_groups(void);
void clear_all_fields(void);

const char *combobox_text(const char *key);
int32_t combobox_find_text(const char *key, const char *text);

void update_lineedit_num(const char *key, EmbReal num, bool useAnglePrecision);
void update_line_edit_str_if_varies(const char *key, const char *str);
void update_lineedit_str(const char *key, const char *str, char *strList[]);
void update_lineedit_bool(const char *key, bool val, bool yesOrNoText);
void update_font_combo_box_str_if_varies(const char *str);

void setHistory(const char *txt);
void obj_set_rect(uint32_t obj, EmbReal x, EmbReal y, EmbReal w, EmbReal h);

void combobox_selection_index_changed(const char *key, int32_t setting_key,
    int32_t index, uint32_t defaultColor);

void combo_ruler_metric_index_changed(int);
void combo_qsnap_locator_color_changed(int);
void combo_cool_grip_color_changed(int);
void combobox_hot_grip_color_index_changed(int);

void choose_mdi_bg_logo(void);
void choose_mdi_bg_texture(void);

void prompt_history_appended(char *txt);
void log_prompt_input(char *txt);
void prompt_input_previous(void);
void prompt_input_next(void);

/* ----------------------------- Object Core ------------------------------- */

void free_object(ObjectCore *core);

ObjectCore *obj_get_core(uint32_t id);

uint32_t create_object(int type_, uint32_t rgb);
uint32_t copy_object(uint32_t);

```

```
uint32_t create_arc(EmbArc arc, uint32_t rgb);
uint32_t create_circle(EmbCircle circle, uint32_t rgb);
uint32_t create_dimleader(EmbLine leader, uint32_t rgb);
uint32_t create_ellipse(EmbEllipse ellipse, uint32_t rgb);
uint32_t create_line(EmbLine line, uint32_t rgb);
uint32_t create_path(EmbPath *p, uint32_t rgb);
uint32_t create_point(EmbPoint p, uint32_t rgb);
uint32_t create_polygon(EmbPath *p, uint32_t rgb);
uint32_t create_polyline(EmbPath *path, uint32_t rgb);
uint32_t create_rect(EmbRect rect, uint32_t rgb);
uint32_t create_text_single(char *str, EmbVector v, uint32_t rgb);

EmbVector obj_rubber_point(int32_t id, const char *key);
const char *obj_rubber_text(int32_t id, const char *key);
EmbVector obj_map_rubber(int32_t obj, const char *key);

double obj_length(ObjectCore *obj);

void obj_set_pos(ObjectCore *obj, EmbVector point);
void obj_set_x(ObjectCore *obj, EmbReal x);
void obj_set_y(ObjectCore *obj, EmbReal y);

void obj_set_center(ObjectCore *obj, EmbVector point);
void obj_set_center_x(ObjectCore *obj, EmbReal centerX);
void obj_set_center_y(ObjectCore *obj, EmbReal centerY);

void obj_set_end_point_1(ObjectCore *obj, EmbVector endPt1);
void obj_set_end_point_2(ObjectCore *obj, EmbVector endPt2);

void obj_set_x1(ObjectCore *obj, EmbReal x);
void obj_set_y1(ObjectCore *obj, EmbReal y);
void obj_set_x2(ObjectCore *obj, EmbReal x);
void obj_set_y2(ObjectCore *obj, EmbReal y);

void obj_set_rect(uint32_t obj, EmbReal x, EmbReal y, EmbReal w, EmbReal h);
void obj_set_rotation(uint32_t id, EmbReal rotation);
void obj_set_rubber_mode(uint32_t id, int mode);
void obj_set_rubber_point(uint32_t id, char *key, EmbVector value);
void obj_set_rubber_text(uint32_t id, char *key, char *value);
void obj_vulcanize(int32_t obj);

void obj_calculate_data(uint32_t obj);

EmbVector obj_top_left(ObjectCore *obj);
EmbVector obj_top_right(ObjectCore *obj);
EmbVector obj_bottom_left(ObjectCore *obj);
EmbVector obj_bottom_right(ObjectCore *obj);

EmbRect obj_rect(ObjectCore *obj);

void obj_set_text(ObjectCore *obj, const char *str);
void obj_set_text_font(ObjectCore *obj, const char *font);
void obj_set_text_justify(ObjectCore *obj, const char *justify);
void obj_set_text_size(ObjectCore *obj, EmbReal size);
void obj_set_text_style(ObjectCore *obj, bool bold, bool italic, bool under, bool strike, bool over);
void obj_set_text_bold(ObjectCore *obj, bool val);
void obj_set_text_italic(ObjectCore *obj, bool val);
void obj_set_text_underline(ObjectCore *obj, bool val);
void obj_set_text_strikeout(ObjectCore *obj, bool val);
void obj_set_text_overline(ObjectCore *obj, bool val);
void obj_set_text_backward(ObjectCore *obj, bool val);
void obj_set_text_upside_down(ObjectCore *obj, bool val);

void obj_move_by(int32_t id, EmbVector delta);
int32_t obj_find_index(int32_t id, EmbVector delta);
void obj_update_rubber_grip(uint32_t obj);

EmbVectorList *all_grip_points(int32_t obj_id);
EmbVector mouse_snap_point(int32_t obj_id, EmbVector mousePoint);

int find_widget_list(const char *key);
char *qcolor_from_uint32_t(uint32_t color);
ScriptValue allow_rubber_command(ScriptEnv *context);

/* ---------------------- Script Environment management -------------------- */
int load_global_state(ScriptValue *root, char *asset_dir);

ScriptValue *emb_create_root(void);
int emb_create_leaf(ScriptValue *branch, int type, char *label, char *data);
ScriptValue *emb_get_var(ScriptValue *node, const char *key);
const char *emb_get_str(ScriptValue *node, const char *key);
void emb_free_root(ScriptValue *root);
void emb_print_tree(ScriptValue *tree, int indent);

/* ---------------------------- Global Variables --------------------------- */
/* Global variables and constants we need to access anywhere in the program
* with minimal overhead.
```

```
*/

extern Widget widget_list[MAX_WIDGETS];
extern CommandData command_data[MAX_COMMANDS];
extern StringMap aliases[MAX_ALIASES];
extern Setting setting[N_SETTINGS];
extern GroupBoxData group_box_list[];
extern Translation translations[];
extern Design designs[];
extern const char *svgs[];
extern ScriptValue *root;

extern const char *config_table[];
extern char *settings_data[];

extern ScriptValue *config;
extern int n_variables;

extern const char *index_th_name[];
extern bool exitApp;

extern const char *circle_origin_path;

/* Menus */
extern char *menu_list[];
extern int menubar_full_list[];
extern int menubar_no_docs[];

extern char *file_menu[];
extern char *edit_menu[];
extern char *view_menu[];
extern char *zoom_menu[];
extern char *pan_menu[];
extern char *help_menu[];
extern char *draw_menu[];
extern char *tools_menu[];
extern char *modify_menu[];
extern char *dimension_menu[];
extern char *sandbox_menu[];
extern char *recent_menu[];
extern char *window_menu[];

/* Toolbars */
extern char *toolbar_list[];
extern int toolbars_when_docs[];

extern int top_toolbar[];
extern int left_toolbar[];
extern int bottom_toolbar[];
extern int toolbar_horizontal[];

extern char *file_toolbar[];
extern char *edit_toolbar[];
extern char *view_toolbar[];
extern char *zoom_toolbar[];
extern char *pan_toolbar[];
extern char *icon_toolbar[];
extern char *help_toolbar[];
extern char *draw_toolbar[];
extern char *inquiry_toolbar[];
extern char *modify_toolbar[];
extern char *dimension_toolbar[];
extern char *sandbox_toolbar[];
extern char *layer_toolbar[];
extern char *properties_toolbar[];
extern char *text_toolbar[];
extern char *prompt_toolbar[];

/* Widget Groups */
extern char *grid_load_from_file_group[];
extern char *defined_origin_group[];
extern char *rectangular_grid_group[];
extern char *circular_grid_group[];
extern char *center_on_origin_group[];
extern char *grid_enabled_group[];
extern char *rectangular_grid_visible_group[];
extern char *circular_grid_visible_group[];

/* Settings */
extern char *settings_tab_labels[];
extern int preview_to_dialog[];
extern int accept_to_dialog[];
extern int render_hints[];

/* Objects */
extern char *object_names[];

```

```
/* Testing */
extern char *coverage_test[];

/* Misc */
extern char *tips[];
extern char *extensions[];

extern bool document_memory[MAX_OPEN_FILES];

extern char formatFilterOpen[MAX_LONG_STRING];
extern char formatFilterSave[MAX_LONG_STRING];
extern char open_files_path[MAX_STRING_LENGTH];
extern char prompt_color_[MAX_STRING_LENGTH];
extern char prompt_selection_bg_color_[MAX_STRING_LENGTH];
extern char prompt_bg_color_[MAX_STRING_LENGTH];
extern char prompt_selection_color_[MAX_STRING_LENGTH];

extern char lastCmd[MAX_STRING_LENGTH];

extern ScriptEnv *global;

extern const ScriptValue script_null;
extern const ScriptValue script_true;
extern const ScriptValue script_false;

extern bool blinkState;

extern int testing_mode;
extern char *button_list[];

extern char *_appName_;
extern char *_appVer_;
extern char settings_file[MAX_STRING_LENGTH];

extern char *layer_list[];
extern char *color_list[];
extern char *line_type_list[];
extern char *line_weight_list[];
extern char *text_size_list[];

extern char *geometry_type_keys[];

extern char *editor_list[];
extern char *combobox_list[];

extern char *grid_layout[];
extern char *zoom_layout[];

extern char **xpm_icons[];
extern char *xpm_icon_labels[];

extern char recent_files[MAX_FILES][MAX_STRING_LENGTH];

extern int numOfDocs;
extern int docIndex;
extern bool shiftKeyPressedState;
extern bool cmdActive;
extern bool rapidFireEnabled;
extern bool isBlinking;
extern bool key_state[N_KEY_SEQUENCES];

extern StringMap aliasHash[MAX_ALIASES];
extern IntMap obj_index[MAX_OBJECTS];

extern EmbIdList *cutCopyObjectList;

extern bool blinkState;

extern int iconSize;

extern bool pickAdd;

extern int promptInputNum;

extern int precisionAngle;
extern int precisionLength;

extern int n_aliases;
extern int n_objects;
extern int n_commands;
extern int n_widgets;
extern int n_actions;

extern char curText[MAX_STRING_LENGTH];

extern char promptHistoryData[MAX_LONG_STRING];

```

```
extern char defaultPrefix[200];
extern char prefix[200];
extern char curCmd[200];

/* Used when checking if fields vary.  */
extern char fieldOldText[MAX_STRING_LENGTH];
extern char fieldNewText[MAX_STRING_LENGTH];
extern char fieldVariesText[MAX_STRING_LENGTH];
extern char fieldYesText[MAX_STRING_LENGTH];
extern char fieldNoText[MAX_STRING_LENGTH];
extern char fieldOnText[MAX_STRING_LENGTH];
extern char fieldOffText[MAX_STRING_LENGTH];

extern WidgetData grid_gb_data[];
extern WidgetData zoom_gb_data[];

#ifdef __cplusplus
}
#endif

#endif

```

25.2 src/interface.c File Reference

```
#include <stdio.h>
#include <string.h>
#include <assert.h>
#include <stdlib.h>
#include <limits.h>
#include <errno.h>
#include <time.h>
#include <math.h>
#include <X11/X.h>
#include <X11/Xlib.h>
#include <X11/Xatom.h>
#include "glad/glad.h"
#include <GL/gl.h>
#include <GL/glx.h>
#include <GL/glu.h>
#include <GLFW/glfw3.h>
#include "nanovg.h"
#include "nanovg_gl.h"
#include "nanosvg.h"
#include "core.h"
```

Macros

- #define **NANOVG_GLES2_IMPLEMENTATION**
- #define **NANOSVG_IMPLEMENTATION**
- #define **MAX_HISTORY** 300

Functions

- EmbVector **doc_map_to_scene** (int32_t doc, EmbVector v)
- GLFWwindow ∗ **create_window** (int32_t width, int32_t height, const char ∗title)
- void **draw_rect** (NVGcontext ∗vg, EmbRect rect, EmbColor color)
- void **draw_button** (NVGcontext ∗vg, Button button, float ∗bounds)
- void **draw_text** (NVGcontext ∗vg, int x, int y, char ∗font, char ∗txt, EmbColor color, float ∗bounds)
- void **draw_view** (NVGcontext ∗vg, EmbRect container)
- void **draw_mdiarea** (NVGcontext ∗vg, EmbRect container)
- void **draw_dockarea** (NVGcontext ∗vg, EmbRect container)
- void **draw_nvg_toolbar** (NVGcontext ∗vg, EmbRect container)
- void **draw_prompt** (NVGcontext ∗vg, EmbRect container)

- void **draw_statusbar** (NVGcontext ∗vg, EmbRect container)
- int **main** (int argc, char ∗argv[ ])
- uint32_t **RGB** (uint8_t r, uint8_t g, uint8_t b)
- void **update_all_background_color** (uint32_t color)
- void **preview_update** (void)
- void **update_view** (void)
- void **mdiarea_set_bg** (uint32_t color)
- Button **make_menubar_button** (NVGcontext ∗vg, int x, int y, char ∗text, float ∗bounds)
- Button **make_toolbar_button** (int x, int y, char ∗text)
- void **draw_menubar** (NVGcontext ∗vg, EmbRect container)
- void **svg_test** (void)
- void **draw_interface** (NVGcontext ∗vg)
- int **start_frame** (GLFWwindow ∗window, EmbReal ∗prevt)
- void **end_frame** (GLFWwindow ∗window)
- void **messagebox** (const char ∗logo, const char ∗title, const char ∗text)
- int32_t **free_glfw** (void)
- int **glfw_application** (int argc, char ∗argv[ ])
- void **settings_prompt** (void)
- void **accept_interface_color** (int32_t key, uint32_t color)
- void **preview_interface_color** (int32_t key, uint32_t color)
- void **dialog_interface_color** (int32_t key, uint32_t color)
- void **prompt_history_appended** (char ∗txt)
- void **log_prompt_input** (char ∗txt)
- void **prompt_input_previous** (void)
- void **prompt_input_next** (void)
- EmbVectorList ∗ **create_vector_list** (void)
- void **append_vector_to_list** (EmbVectorList ∗list, EmbVector v)
- void **remove_vector_from_list** (EmbVectorList ∗list, int32_t position)
- void **free_vector_list** (EmbVectorList ∗v)
- EmbIdList ∗ **create_id_list** (void)
- void **append_id_to_list** (EmbIdList ∗list, int32_t i)
- bool **id_list_contains** (EmbIdList ∗list, int32_t a)
- void **remove_id_from_list** (EmbIdList ∗list, int32_t position)
- void **free_id_list** (EmbIdList ∗list)
- int **find_int_map** (IntMap ∗map, int key)
- int **find_in_map** (StringMap ∗hash, int length, const char ∗key)
- void **nanosleep_** (int time)
- void **run_testing** (void)
- ScriptValue **test_command** (ScriptEnv ∗context)
- int **argument_checks** (ScriptEnv ∗context, int id)
- char ∗ **translate** (const char ∗msg)
- ScriptEnv ∗ **create_script_env** ()
- void **free_script_env** (ScriptEnv ∗context)
- ScriptEnv ∗ **pack** (ScriptEnv ∗context, const char ∗fmt,...)
- ScriptValue **call** (ScriptEnv ∗context, char ∗cmd,...)
- int **config_find** (const char ∗key)
- char ∗ **config_str** (const char ∗key)
- void **load_command_data** (char ∗label)
- int **load_data** (void)
- int **load_settings** (const char ∗appDir, const char ∗fname)
- int **save_settings** (char ∗appDir, char ∗fname)
- int **get_command_id** (char ∗name)
- bool **pattern_save** (EmbPattern ∗pattern, char ∗fileName)
- ScriptValue ∗ **setting_ptr** (int key, int mode)
- void **copy_setting** (int key, int dst, int src)

- const char ∗ **platform_string** (void)
- void **update_lineedit_vector** (EmbGeometry ∗g, const char ∗key, int id, bool editable)
- void **update_editors** (int32_t obj)
- void **edit_field** (int32_t id, const char ∗label, const char ∗text)
- void **show_groups** (int objType)
- void **spin_box_display_select_box_alpha_changed** (int value)
- void **combo_box_prompt_font_family_changed** (char ∗family)
- void **combo_box_prompt_font_style_changed** (char ∗style)
- void **spin_box_prompt_font_size_changed** (int value)
- void **check_box_lwt_real_render_changed** (int checked)
- void **hide_all_groups** (void)
- void **clear_all_fields** (void)
- void **checkBoxGridCenterOnOriginStateChanged** (int checked)
- void **checkBoxLwtShowLwtStateChanged** (int checked)
- void **accept_settings** (void)
- void **comboBoxGridTypeCurrentIndexChanged** (const char ∗type)
- void **checkBoxGridColorMatchCrossHairStateChanged** (int checked)
- void **checkBoxGridLoadFromFileStateChanged** (int checked)
- void **doc_init** (int32_t doc)
- void **free_doc** (int32_t doc)
- void **doc_clear_selection** (int32_t doc)
- void **doc_zoom_window** (int32_t doc)
- void **doc_preview_on** (int32_t doc, int clone, int mode, EmbReal x, EmbReal y, EmbReal data_)
- void **doc_preview_off** (int32_t doc)
- void **doc_spare_rubber** (int32_t doc, int64_t id)
- void **update_color_linetype_lineweight** (void)
- void **doc_cut** (int32_t doc)
- void **doc_create_grid** (int32_t doc, const char ∗gridType)
- void **open_recent_file** (void)
- void **statusbar_toggle** (const char ∗key, bool on)
- void **doc_copy** (int32_t doc)
- void **doc_set_rubber_mode** (int32_t doc, int mode)
- void **doc_set_rubber_point** (int32_t doc, char ∗key, EmbVector point)
- void **doc_set_rubber_text** (int32_t doc, char ∗key, EmbString txt)
- void **copy_object_list** (EmbIdList ∗dst, EmbIdList ∗src)
- void **doc_copy_selected** (int32_t doc)
- void **doc_vulcanize_rubber_room** (int32_t doc)
- void **doc_center_at** (int32_t doc, EmbVector centerPoint)
- bool **doc_allow_zoom_in** (int32_t doc)
- bool **doc_allow_zoom_out** (int32_t doc)
- bool **doc_allow_rubber** (int32_t doc)
- ScriptValue ∗ **doc_get_attr** (int32_t doc, uint32_t key)

  Helper function to gain access to the document level attributes.
- bool **doc_get_bool** (int32_t doc, uint32_t key)

  Helper function to gain access to the document level boolean attribute.
- int32_t **doc_get_int** (int32_t doc, uint32_t key)

  Helper function to gain access to the document level integer attribute.
- void **doc_set_attr** (int32_t doc, uint32_t key, ScriptValue value)

  Helper function to set a document level attribute.
- void **doc_set_bool** (int32_t doc, uint32_t key, bool value)

  Helper function to set a document-level boolean attribute.
- void **doc_set_int** (int32_t doc, uint32_t key, int32_t value)

  Helper function to set a document-level integer attribute.
- void **doc_set_grid_color** (int32_t doc, uint32_t color)

- void **doc_set_ruler_color** (int32_t doc, uint32_t color)
- void **doc_set_cross_hair_color** (int32_t doc, uint32_t color)
- void **doc_toggle** (int32_t doc, const char ∗key, bool on)
- void **doc_stop** (int32_t doc)
- void **doc_delete_pressed** (int32_t doc)
- void **doc_escape_pressed** (int32_t doc)
- void **doc_delete_selected** (int32_t doc)
- void **doc_move_selected** (int32_t doc, EmbVector delta)
- void **doc_rotate_selected** (int32_t doc, EmbVector v, EmbReal rot)
- void **doc_mirror_selected** (int32_t doc, EmbReal x1, EmbReal y1, EmbReal x2, EmbReal y2)
- void **doc_scale_selected** (int32_t doc, EmbReal x, EmbReal y, EmbReal factor)
- int **string_replace** (char ∗result, char ∗base, char ∗from, char ∗to)
- ViewData ∗ **create_view_data** (int32_t doc, int width, int height, int unit)
- void **free_view_data** (ViewData ∗vdata)
- void **add_line** (ViewData ∗vdata, EmbVector start, EmbVector end)
- int **create_ruler_lines** (ViewData ∗vdata)
- void **set_visibility_group** (char ∗keylist[ ], bool visibility)
- void **set_enabled_group** (char ∗keylist[ ], bool enabled)
- void **run_command_main** (const char ∗cmd)
- void **run_command_click** (const char ∗cmd, EmbReal x, EmbReal y)
- void **run_command_move** (const char ∗cmd, EmbReal x, EmbReal y)
- void **run_command_context** (const char ∗cmd, const char ∗str)
- void **run_command_prompt** (const char ∗cmd)
- ScriptValue **run_cmd** (ScriptEnv ∗context, const char ∗cmd)
- void **draw_frame** (void)

Variables

- char ∗ objectTextJustifyList [ ]
- const char ∗ help_msg
- char ∗ ruler_svg
- double **zoomInLimit** = 1.0e-10
- double **zoomOutLimit** = 1.0e10
- EmbString **prompt_input_list** [MAX_HISTORY]
- int **prompt_history_size** = 0
- int **prompt_history_position** = 0
- NVGcontext ∗ **vg**
- int **icon_size** = 16
- Button **menu_buttons** [10]
- Button **tool_buttons_a** [100]
- const char ∗ **sans_font** = "/usr/share/fonts/truetype/open-sans/OpenSans-Regular.ttf"
- const char ∗ **mono_font** = "/usr/share/fonts/truetype/freefont/FreeMono.ttf"
- const char ∗ **icon_font** = "/usr/share/fonts/truetype/open-sans/OpenSans-Regular.ttf"
- EmbColor **toolbar_bg_color** = {.r=128, .g=128, .b=128}
- EmbColor **toolbar_button_color** = {.r=208, .g=208, .b=208}
- EmbColor **toolbar_text** = {.r=0, .g=0, .b=0}
- EmbColor **prompt_bg_color** = {.r=255, .g=255, .b=255}
- EmbColor **prompt_color** = {.r=0, .g=0, .b=0}
- EmbColor **scrollbar_color** = {.r=100, .g=100, .b=100}
- int **window_width** = 640
- int **window_height** = 480
- int **toolbar_height** = 40
- float **button_padding** = 5.0
- float **prompt_divider** = 405.0F
- float **statusbar_depth** = 40.0F
- char prompt_text [50][MAX_STRING_LENGTH]
- int **n_prompt_lines** = 3
- const char ∗ config_table [ ]

25.2.1 Detailed Description

Embroidermodder 2 (GLFW version)
Copyright 2011-2025 The Embroidermodder Team Embroidermodder 2 is Open Source Software, see LICENSE.md for licensing terms.
Read the reference manual (https://www.libembroidery.org/downloads/emrm.pdf) for advice on altering this file.
Main window wrangling and drawing functions.
Operating system-specific system calls like usleep go in this file: see the block of system-specific headers below.

25.2.2 Variable Documentation

25.2.2.1 config_table `const char* config_table[]`

Initial value:

```
= {
    END_SYMBOL
}
```

25.2.2.2 help_msg `const char* help_msg`

Initial value:

```
=
     " ___ _____ ___  ___   __  _ ___  ___ ___   _____  __  ___  ___  ___ ___     ___ \n"
    "| __|     | _ \\| _ \\ /  \\| |   \\| __| _ \\ |     |/  \\|   \\|   \\| __| _ \\  |__ \\\n"
    "| __| | | | _ <|   /| () | | |) | __|   / | | | | () | |) | |) | __|   /  / __/\n"
    "|___|_|_|_|___/|_|\\_\\\\\__/|_|___/|___|_|\\_\\|_|_|_|\\__/|___/|___/|___|_|\\_\\ |___|\n"
    " _____________________________________________________________________________ \n"
    "|                                                                             |\n"
    "|                   https://www.libembroidery.org                             |\n"
    "|_____________________________________________________________________________|\n"
    "\n"
    "Usage:  embroidermodder [options] files ...\n"
    "Options:\n"
    "  -d, --debug      Print lots of debugging information.\n"
    "  -h, --help       Print this message and exit.\n"
    "  -v, --version    Print the version number of embroidermodder and exit.\n"
    "\n"
```

25.2.2.3 objectTextJustifyList `char* objectTextJustifyList[]`

Initial value:

```
= {
    "Left",
    "Center",
    "Right",
    "Middle",
    "Top Left",
    "Top Center",
    "Top Right",
    "Middle Left",
    "Middle Center",
    "Middle Right",
    "Bottom Left",
    "Bottom Center",
    "Bottom Right",
    "END"
}
```

25.2.2.4 prompt_text `char prompt_text[50][MAX_STRING_LENGTH]`

Initial value:

```
= {
    "prompt> ",
    "",
    "",
    ""
}
```

25.2.2.5 ruler_svg `char* ruler_svg`

Initial value:

```
= "<svg width=\"WIDTH\" height=\"HEIGHT\" viewBox=\"0 0 WIDTH HEIGHT\" fill=\"none\"
      xmlns=\"http://www.w3.org/2000/svg\">"
    "<rect x=\"0\" y=\"0\" width=\"WIDTH\" height=\"HEIGHT\" fill=\"RULER_COLOR\" stroke=\"RULER_BORDER\"
      stroke-width=\"1\" />"
    "</svg>"
```

25.3 src/objects.c File Reference

```
#include <stdio.h>
#include <string.h>
#include "core.h"
```

Functions

- uint32_t **copy_object** (uint32_t obj)
- void **free_object** (ObjectCore *core)
- uint32_t **create_arc** (EmbArc arc, uint32_t rgb)
- uint32_t **create_circle** (EmbCircle circle, uint32_t rgb)
- uint32_t **create_dimleader** (EmbLine line, uint32_t rgb)
- uint32_t **create_ellipse** (EmbEllipse ellipse, uint32_t rgb)
- uint32_t **create_image** (EmbRect rect, uint32_t rgb)
- uint32_t **create_line** (EmbLine line, uint32_t rgb)
- uint32_t **create_path** (EmbPath *p, uint32_t rgb)
- uint32_t **create_polygon** (EmbPath *p, uint32_t rgb)
- uint32_t **create_polyline** (EmbPath *path, uint32_t rgb)
- uint32_t **create_point** (EmbPoint point, uint32_t rgb)
- uint32_t **create_rect** (EmbRect rect, uint32_t rgb)
- uint32_t **create_text_single** (EmbString str, EmbVector v, uint32_t rgb)
- EmbVector **find_mouse_snap_point** (EmbVectorList *snap_points, EmbVector mouse_point)
- EmbVector **obj_delta** (ObjectCore *obj)
- void **obj_set_pos** (ObjectCore *obj, EmbVector point)
- void **obj_set_rubber_mode** (uint32_t id, int mode)
- void **obj_set_x** (ObjectCore *obj, EmbReal x)
- void **obj_set_y** (ObjectCore *obj, EmbReal y)
- void **obj_set_center** (ObjectCore *obj, EmbVector center)
- void **obj_set_center_x** (ObjectCore *obj, EmbReal centerX)
- void **obj_set_center_y** (ObjectCore *obj, EmbReal centerY)
- double **obj_length** (ObjectCore *obj)
- void **obj_set_x1** (ObjectCore *obj, EmbReal x)
- void **obj_set_y1** (ObjectCore *obj, EmbReal y)
- void **obj_set_x2** (ObjectCore *obj, EmbReal x)
- void **obj_set_y2** (ObjectCore *obj, EmbReal y)
- ScriptValue **arc_command** (ScriptEnv *context)
- ScriptValue **circle_command** (ScriptEnv *context)
- ScriptValue **circle_click** (ScriptEnv *context)
- ScriptValue **circle_prompt** (ScriptEnv *context)
- ScriptValue **ellipse_command** (ScriptEnv *context)
- ScriptValue **line_command** (ScriptEnv *context)
- ScriptValue **path_command** (ScriptEnv *context)
- ScriptValue **point_command** (ScriptEnv *context)
- ScriptValue **polygon_command** (ScriptEnv *context)
- ScriptValue **polyline_command** (ScriptEnv *context)
- ScriptValue **rectangle_command** (ScriptEnv *context)
- ScriptValue **singlelinetext_command** (ScriptEnv *)

- ScriptValue **quickleader_command** (ScriptEnv ∗context)
- EmbVector **obj_top_left** (ObjectCore ∗obj)
- EmbVector **obj_top_right** (ObjectCore ∗obj)
- EmbVector **obj_bottom_left** (ObjectCore ∗obj)
- EmbVector **obj_bottom_right** (ObjectCore ∗obj)
- void **obj_set_text_font** (ObjectCore ∗obj, const char ∗font)
- void **obj_set_text_justify** (ObjectCore ∗obj, const char ∗justify)
- void **obj_set_text_size** (ObjectCore ∗obj, EmbReal size)
- void **obj_set_text_style** (ObjectCore ∗obj, bool bold, bool italic, bool under, bool strike, bool over)
- void **obj_set_text_bold** (ObjectCore ∗obj, bool val)
- void **obj_set_text_italic** (ObjectCore ∗obj, bool val)
- void **obj_set_text_underline** (ObjectCore ∗obj, bool val)
- void **obj_set_text_strikeout** (ObjectCore ∗obj, bool val)
- void **obj_set_text_overline** (ObjectCore ∗obj, bool val)
- void **obj_set_text_backward** (ObjectCore ∗obj, bool val)
- void **obj_set_text_upside_down** (ObjectCore ∗obj, bool val)
- void **obj_vulcanize** (int32_t obj)
- void **obj_set_rubber_point** (uint32_t id, EmbString key, EmbVector point)
- void **obj_set_rubber_text** (uint32_t id, EmbString key, EmbString txt)
- EmbVector **obj_rubber_point** (int32_t id, const char ∗key)
- const char ∗ **obj_rubber_text** (int32_t id, const char ∗key)
- EmbArc **emb_arc_set_radius** (EmbArc arc, EmbReal radius)
- void **obj_grip_edit** (int32_t obj_id, EmbVector before, EmbVector after)
- EmbVectorList ∗ **all_grip_points** (int32_t obj_id)
- EmbVector **mouse_snap_point** (int32_t obj_id, EmbVector mousePoint)
- ScriptValue **add_arc_command** (ScriptEnv ∗context)
- ScriptValue **add_circle_command** (ScriptEnv ∗context)
- ScriptValue **add_dimleader_command** (ScriptEnv ∗context)
- ScriptValue **add_ellipse_command** (ScriptEnv ∗context)
- ScriptValue **add_line_command** (ScriptEnv ∗context)
- ScriptValue **add_infinite_line_command** (ScriptEnv ∗context)
- ScriptValue **add_ray_command** (ScriptEnv ∗context)
- ScriptValue **add_triangle_command** (ScriptEnv ∗context)
- ScriptValue **add_rounded_rectangle_command** (ScriptEnv ∗context)
- ScriptValue **add_regular_polygon_command** (ScriptEnv ∗context)
- ScriptValue **add_horizontal_dimension_command** (ScriptEnv ∗context)
- ScriptValue **add_vertical_dimension_command** (ScriptEnv ∗context)
- ScriptValue **add_polygon_command** (ScriptEnv ∗context)
- ScriptValue **add_polyline_command** (ScriptEnv ∗context)
- ScriptValue **add_point_command** (ScriptEnv ∗context)
- ScriptValue **add_rectangle_command** (ScriptEnv ∗context)
- ScriptValue **add_slot_command** (ScriptEnv ∗context)
- ScriptValue **add_textsingle_command** (ScriptEnv ∗context)
- ScriptValue **add_textmulti_command** (ScriptEnv ∗context)

Variables

- const char ∗ **justify_options** [ ]

25.3.1 Detailed Description

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Read the reference manual (https://www.libembroidery.org/downloads/emrm.pdf) for advice on altering this file.
ObjectCore: the additional information and tools on top of EmbGeometry required for the user interface.
Geometry Commands for user interface (internal functions only are in "objects.c")
Rubber Point and Grip Point Management

25.4 src/qt.h File Reference

```
#include "core.h"
#include <QAction>
#include <QApplication>
#include <QGraphicsPathItem>
#include <QtPrintSupport>
```

Data Structures

- class Object
- class SelectBox
- class LayerManager
- class MdiArea
- class Document
- class UndoableCommand
- class UndoEditor
- class PropertyEditor
- class MdiWindow
- class Application
- class CmdPromptInput
- class MainWindow

Variables

- Document * **documents** [MAX_OPEN_FILES]

25.4.1 Detailed Description

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Read the reference manual (https://www.libembroidery.org/downloads/emrm.pdf) for advice on altering this file.
Qt wrapper

25.5 qt.h

Go to the documentation of this file.

```
1
15 #ifndef EMBROIDERMODDER_GUI_HEADER
16 #define EMBROIDERMODDER_GUI_HEADER
17
18 #include "core.h"
19
20 #include <QAction>
21 #include <QApplication>
22 #include <QGraphicsPathItem>
23 #include <QtPrintSupport>
24
```

```
class Object:  public QGraphicsPathItem
{
public:
    ObjectCore *core;

    QGraphicsPathItem path_;
    QPen objPen;
    QPen lwtPen;
    QLineF objLine;
    QPainter* obj_painter;

    QPainterPath textPath;
    QPainterPath lineStylePath;
    QPainterPath arrowStylePath;
    QPainterPath normalPath;

    Object(int type_, QRgb rgb, Qt::PenStyle lineType, QGraphicsItem* item = 0);
    ~Object() { free_object(core); }

    void gripEdit(EmbVector before, EmbVector after);
    QPainterPath shape()const { return path(); }

    void drawRubberLine(QLineF rubLine, QPainter* painter = 0, const char* colorFromScene = 0);

    void setObjectSize(double width, EmbReal height);

    QPainterPath objectCopyPath() const;
    QPainterPath objectSavePath() const;
    QList<QPainterPath> objectSavePathList()const { return subPathList(); }
    QList<QPainterPath> subPathList() const;

protected:
    void paint(QPainter*, const QStyleOptionGraphicsItem*, QWidget*);
};

class SelectBox:  public QRubberBand
{
    Q_OBJECT

public:
    SelectBox(Shape s, QWidget* parent = 0);

public slots:
    void setDirection(int dir);
    void setColors(const QColor& colorL, const QColor& fillL, const QColor& colorR, const QColor& fillR,
   int newAlpha);

protected:
    void paintEvent(QPaintEvent*);

private:
    QColor leftBrushColor;
    QColor rightBrushColor;
    QColor leftPenColor;
    QColor rightPenColor;
    uint8_t alpha;

    QBrush dirBrush;
    QBrush leftBrush;
    QBrush rightBrush;

    QPen dirPen;
    QPen leftPen;
    QPen rightPen;

    bool boxDir;

    void forceRepaint();
};

class LayerManager:  public QDialog
{
    Q_OBJECT

public:
    LayerManager(QWidget *parent = 0);
     ~LayerManager() { restore_cursor(); }
     LayerData data[MAX_LAYERS];

     void addLayer(char *name, bool visible, bool frozen, EmbReal zValue,
         QRgb color, char *lineType, char *lineWeight, bool print);

     QStandardItemModel* layerModel;
     QSortFilterProxyModel* layerModelSorted;
     QTreeView* treeView;
};

```

```
class MdiArea:  public QMdiArea
{
    Q_OBJECT

public:
    MdiArea(QWidget* parent = 0);
    ~MdiArea() {}

    bool useLogo;
    bool useTexture;
    bool useColor;

    QPixmap bgLogo;
    QPixmap bgTexture;
    QColor bgColor;

    void forceRepaint();

public slots:
    void cascade();
    void tile();
protected:
    virtual void mouseDoubleClickEvent(QMouseEvent* e);
    virtual void paintEvent(QPaintEvent* e);
};

/* .  */
class Document:  public QGraphicsView
{
    Q_OBJECT

public:
    Document(QGraphicsScene* theScene, QWidget* parent);
    ~Document() { free_doc(data->id); }

    DocumentData *data;

    QVector<int64_t> hashDeletedObjects;
    QPainterPath gridPath;
    QPainterPath originPath;

    QPainter* scene_painter;

    Object* gripBaseObj;

    QGraphicsScene* gscene;
    QUndoStack* undoStack;

    SelectBox* selectBox;
    QGraphicsItemGroup* pasteObjectItemGroup;

protected:
    void mouseDoubleClickEvent(QMouseEvent* event);
    void mousePressEvent(QMouseEvent* event);
    void mouseMoveEvent(QMouseEvent* event);
    void mouseReleaseEvent(QMouseEvent* event);
    void wheelEvent(QWheelEvent* event);
    void contextMenuEvent(QContextMenuEvent* event);
    void drawBackground(QPainter* painter, const QRectF& rect);
    void draw_rulers(QPainter* painter, const QRectF& rect);
    void drawForeground(QPainter* painter, const QRectF& rect);
    void enterEvent(QEvent* event);
};

extern Document *documents[MAX_OPEN_FILES];

class UndoableCommand :  public QUndoCommand
{
public:
    UndoableCommand(int type_, QString text, Object* obj, int32_t v,
        QUndoCommand* parent = 0);
    UndoableCommand(int type_, EmbVector delta, QString text, Object* obj,
        int32_t v, QUndoCommand* parent = 0);
    UndoableCommand(int type_, EmbVector pivot, EmbReal rotAngle, QString text,
        Object* obj, int32_t v, QUndoCommand* parent = 0);
    UndoableCommand(int type_, QString type, int32_t v, QUndoCommand* parent = 0);
    UndoableCommand(int type_, EmbVector start, EmbVector end, QString text,
        Object* obj, int32_t v, QUndoCommand* parent = 0);

    void undo();
    void redo();
    void rotate(double x, EmbReal y, EmbReal rot);
    int id()const { return 1234; }
    bool mergeWith(const QUndoCommand* command);
    void mirror();

    UndoData data;
```

```
    Object *object;
    QTransform toTransform;
    QTransform fromTransform;
};

class UndoEditor :  public QDockWidget
{
    Q_OBJECT

public:
    UndoEditor(QString iconDirectory = "", QWidget* widgetToFocus = 0,
        QWidget* parent = 0); /*, Qt::WindowFlags flags = 0); */
    ~UndoEditor() {}

    void addStack(QUndoStack* stack);

    bool canUndo() const;
    bool canRedo() const;

    QString undoText() const;
    QString redoText() const;

public slots:
    void undo();
    void redo();

    void updateCleanIcon(bool opened);

private:
    QWidget* focusWidget;

    QString iconDir;
    int iconSize;

    QUndoGroup* undoGroup;
    QUndoView*  undoView;
};

class PropertyEditor:  public QDockWidget
{
    Q_OBJECT

public:
    PropertyEditor(QString iconDirectory = "", bool pickAddMode = true,
        QWidget* widgetToFocus = 0, QWidget* parent = 0);
        /*, Qt::WindowFlags flags = 0); */
    ~PropertyEditor() {}

    void togglePickAddMode();

protected:
    bool eventFilter(QObject *obj, QEvent *event);

signals:
    void pick_add_mode_toggled();

public slots:
    void setSelectedItems(QList<QGraphicsItem*> itemList);
    void update_pick_add_modeButton(bool pickAddMode);
};

class MdiWindow:  public QMdiSubWindow
{
    Q_OBJECT

public:
    MdiWindow(const int theIndex, QMdiArea* parent, Qt::WindowFlags wflags);
    ~MdiWindow() {
        debug_message("MdiWindow Destructor()");
        emb_pattern_free(documents[doc_index]->data->pattern);
    }

    QMdiArea* mdiArea;
    QGraphicsScene* gscene;
    int32_t doc_index;

    QPrinter printer;

    virtual QSize sizeHint() const;
    QString getShortCurrentFile();
    bool loadFile(char *fileName);
signals:
    void sendCloseMdiWin(MdiWindow*);

public slots:
    void closeEvent(QCloseEvent* e);
    void onWindowActivated();
```

```

    void print();
    void saveBMC();

private:
    void setCurrentFile(char *fileName);
};

/* On Mac, if the user drops a file on the app's Dock icon, or uses Open As,
* then this is how the app actually opens the file.
*/
class Application :  public QApplication
{
    Q_OBJECT
public:
    Application(int argc, char **argv);
protected:
    virtual bool event(QEvent *e);
};

class CmdPromptInput:  public QLineEdit
{
    Q_OBJECT

public:
    CmdPromptInput(QWidget* parent = 0);
    ~CmdPromptInput() {}

protected:
    void contextMenuEvent(QContextMenuEvent *event);
    bool eventFilter(QObject *obj, QEvent *event);

public slots:
    void checkSelection();
    void updateCurrentText(QString txt);
    void checkEditedText(QString txt);
    void checkChangedText(QString txt);
    void checkCursorPosition(int oldpos, int newpos);
private slots:
    void copyClip();
    void pasteClip();
};

class MainWindow:  public QMainWindow
{
    Q_OBJECT

public:
    MainWindow();
    ~MainWindow() {
        debug_message("Destructor()");

        /* Prevent memory leaks by deleting any unpasted objects.  */
        free_objects(cutCopyObjectList);
        free_id_list(cutCopyObjectList);
    }

    void add_toolbar_to_window(Qt::ToolBarArea area, int data[]);

public slots:
    void recentMenuAboutToShow();
    void windowMenuActivated(bool checked /*, int id*/);

    void closeToolBar(QAction*);
    void floatingChangedToolBar(bool);

    void runCommand();

    void colorSelectorIndexChanged(int index);

protected:
    virtual void resizeEvent(QResizeEvent*);
    void closeEvent(QCloseEvent *event);
};

#endif

```

25.6 src/state.c File Reference

```
#include "core.h"
#include "toml.h"
#include <string.h>
```

Functions

- int **emb_set_var** (ScriptValue ∗node, const char ∗key, ScriptValue value)
- ScriptValue ∗ **emb_get_var** (ScriptValue ∗node, const char ∗key)
- const char ∗ **emb_get_str** (ScriptValue ∗node, const char ∗key)
- size_t load_file_to_buffer (char ∗fname, char ∗buffer)
- toml_table_t ∗ load_script (char ∗asset_dir, char ∗fname)
- int **load_global_state** (ScriptValue ∗root, char ∗asset_dir)

Variables

- char ∗ **os** = "Unknown Operating System"
- ScriptValue ∗ **root**
- ScriptEnv ∗ **global**
- ScriptValue ∗ **config**
- int **n_variables** = 0
- char **formatFilterOpen** [MAX_LONG_STRING]
- char **formatFilterSave** [MAX_LONG_STRING]
- char **open_files_path** [MAX_STRING_LENGTH]
- char **prompt_color_** [MAX_STRING_LENGTH]
- char **prompt_selection_bg_color_** [MAX_STRING_LENGTH]
- char **prompt_bg_color_** [MAX_STRING_LENGTH]
- char **prompt_selection_color_** [MAX_STRING_LENGTH]
- bool **document_memory** [MAX_OPEN_FILES]
- bool **shiftKeyPressedState** = false
- bool **cmdActive**
- bool **rapidFireEnabled** = false
- bool **isBlinking** = false
- int **numOfDocs** = 0
- int **docIndex** = 0
- char **settings_file** [MAX_STRING_LENGTH] = "settings.ini"
- bool key_state [N_KEY_SEQUENCES]
- Setting **setting** [N_SETTINGS]
- char recent_files [MAX_FILES][MAX_STRING_LENGTH]
- const char ∗ index_name [ ]
- const char ∗ index_th_name [ ]
- const ScriptValue script_null
- const ScriptValue script_true
- const ScriptValue script_false
- char ∗ **_appName_** = "Embroidermodder"
- char ∗ **_appVer_** = "v2.0 alpha"
- bool **exitApp** = false
- int **testing_mode** = 0
- char **defaultPrefix** [200]
- char **prefix** [200]
- char **curCmd** [200]
- char ∗ menu_list [ ]
- int menubar_full_list [ ]
- int menubar_no_docs [ ]
- char ∗ file_menu [ ]
- char ∗ edit_menu [ ]
- char ∗ view_menu [ ]
- char ∗ zoom_menu [ ]
- char ∗ pan_menu [ ]
- char ∗ help_menu [ ]

- char * draw_menu [ ]
- char * tools_menu [ ]
- char * modify_menu [ ]
- char * dimension_menu [ ]
- char * sandbox_menu [ ]
- char * recent_menu [ ]
- char * window_menu [ ]
- char * toolbar_list [ ]
- int top_toolbar [ ]
- int left_toolbar [ ]
- int bottom_toolbar [ ]
- int toolbar_horizontal [ ]
- int toolbars_when_docs [ ]
- char * file_toolbar [ ]
- char * edit_toolbar [ ]
- char * view_toolbar [ ]
- char * icon_toolbar [ ]
- char * help_toolbar [ ]
- char * draw_toolbar [ ]
- char * inquiry_toolbar [ ]
- char * modify_toolbar [ ]
- char * dimension_toolbar [ ]
- char * sandbox_toolbar [ ]
- char * layer_toolbar [ ]
- char * properties_toolbar [ ]
- char * text_toolbar [ ]
- char * prompt_toolbar [ ]
- char * grid_load_from_file_group [ ]
- char * defined_origin_group [ ]
- char * rectangular_grid_group [ ]
- char * circular_grid_group [ ]
- char * center_on_origin_group [ ]
- char * settings_tab_labels [ ]
- int preview_to_dialog [ ]
- int accept_to_dialog [ ]
- int render_hints [ ]
- char * **object_names** [ ]
- char * coverage_test [ ]
- char * tips [ ]
- char * **extensions** [ ]
- char * **editor_list** [ ]
- char * combobox_list [ ]
- char * layer_list [ ]
- char * color_list [ ]
- char * line_type_list [ ]
- char * line_weight_list [ ]
- char * text_size_list [ ]
- StringMap **aliasHash** [MAX_ALIASES]
- IntMap **obj_index** [MAX_OBJECTS]
- EmbIdList * **cutCopyObjectList**
- bool **blinkState**
- int **iconSize**
- bool **pickAdd**
- int **promptInputNum** = 0
- int **precisionAngle**

- int **precisionLength**
- int **n_aliases** = 0
- int **n_objects** = 0
- int **n_widgets** = 0
- int **n_actions** = 0
- EmbString **curText**
- char **promptHistoryData** [MAX_LONG_STRING]
- EmbString **fieldOldText**
- EmbString **fieldNewText**
- EmbString **fieldVariesText**
- EmbString **fieldYesText**
- EmbString **fieldNoText**
- EmbString **fieldOnText**
- EmbString **fieldOffText**
- WidgetData zoom_gb_data [ ]
- WidgetData **grid_gb_data** [ ]
- char ∗ button_list [ ]
- char ∗ ui_labels [ ]
- char ∗ patternDetailsBox [ ]
- char ∗ selectionBoxLayout [ ]
- char ∗ promptColorLayout [ ]
- char ∗ rulerLayout [ ]
- GroupBoxData **group_box_list** [ ]
- char ∗ **settings_data** [ ]
- char ∗ grid_enabled_group [ ]
- char ∗ rectangular_grid_visible_group [ ]
- char ∗ circular_grid_visible_group [ ]
- char ∗ grid_layout [ ]
- char ∗ zoom_layout [ ]
- char **lastCmd** [MAX_STRING_LENGTH] = "help"
- char ∗ geometry_type_keys [ ]
- CommandData **command_data** [MAX_COMMANDS]

25.6.1 Detailed Description

Embroidermodder 2.
Copyright 2011-2025 The Embroidermodder Team Embroidermodder 2 is Open Source Software, see LICENSE.md for licensing terms.
Read the reference manual (https://www.libembroidery.org/downloads/emrm.pdf) for advice on altering this file.

25.6.2 Function Documentation

25.6.2.1 load_file_to_buffer()

```
size_t load_file_to_buffer (
    char * fname,
    char * buffer )
```

25.6.2.2 load_script()

```
toml_table_t * load_script (
    char * asset_dir,
    char * fname )
```

Loads init_script 1 Mb buffer to chain together all of the strings in the init_script string table.
TODO: could reuse buffer safely between script loads.

Todo loading of nested structures like arrays of dicts etc.

25.6.3 Variable Documentation

25.6.3.1 accept_to_dialog `int accept_to_dialog[]`

Initial value:

```
= {
    GENERAL_MDI_BG_LOGO,
    GENERAL_MDI_BG_TEXTURE,
    GENERAL_MDI_BG_COLOR,
    DISPLAY_CROSSHAIR_COLOR,
    DISPLAY_BG_COLOR,
    DISPLAY_SELECTBOX_LEFT_COLOR,
    DISPLAY_SELECTBOX_LEFT_FILL,
    DISPLAY_SELECTBOX_RIGHT_COLOR,
    DISPLAY_SELECTBOX_RIGHT_FILL,
    PROMPT_TEXT_COLOR,
    PROMPT_BG_COLOR,
    RULER_COLOR,
    TERMINATOR_SYMBOL
}
```

25.6.3.2 bottom_toolbar `int bottom_toolbar[]`

Initial value:

```
= {
    TOOLBAR_PROMPT,
    TERMINATOR_SYMBOL
}
```

25.6.3.3 button_list `char* button_list[]`

Initial value:

```
= {
    "SNAP",
    "GRID",
    "RULER",
    "ORTHO",
    "POLAR",
    "QSNAP",
    "QTRACK",
    "LWT",
    END_SYMBOL
}
```

25.6.3.4 center_on_origin_group `char* center_on_origin_group[]`

Initial value:

```
= {
    END_SYMBOL
}
```

25.6.3.5 circular_grid_group `char* circular_grid_group[]`

Initial value:

```
= {
    "labelGridSizeRadius",
    "spinBoxGridSizeRadius",
    "labelGridSpacingRadius",
    "spinBoxGridSpacingRadius",
    "labelGridSpacingAngle",
    "spinBoxGridSpacingAngle",
    END_SYMBOL
}
```

25.6.3.6 circular_grid_visible_group `char* circular_grid_visible_group[]`

Initial value:

```
= {
    "labelGridSizeRadius", "spinBoxGridSizeRadius",
    "labelGridSpacingRadius", "spinBoxGridSpacingRadius",
    "labelGridSpacingAngle", "spinBoxGridSpacingAngle",
    END_SYMBOL
```

```
}
```

25.6.3.7 color_list `char* color_list[]`

Initial value:

```
= {
    "colorbylayer", "ByLayer", "",
    "colorbyblock", "ByBlock", "",
    "colorred", "Red", "255,0,0",
    "coloryellow", "Yellow", "255,255,0",
    "colorgreen", "Green", "0,255,0",
    "colorcyan", "Cyan", "0,255,255",
    "colorblue", "Blue", "0,0,255",
    "colormagenta", "Magenta", "255,0,255",
    "colorwhite", "White", "255,255,255",
    "colorother", "Other...", "",
    END_SYMBOL
}
```

25.6.3.8 combobox_list `char* combobox_list[]`

Initial value:

```
= {
    "GeneralLayer",
    "GeneralColor",
    "GeneralLineType",
    "GeneralLineWeight",
    "ArcClockwise",
    "PathVertexNum",
    "PathClosed",
    "PolylineVertexNum",
    "PolylineClosed",
    "TextSingleJustify",
    "TextSingleBackward",
    "TextSingleUpsideDown",
    END_SYMBOL
}
```

25.6.3.9 coverage_test `char* coverage_test[]`

Initial value:

```
= {
    "new",
    "icon16",
    "icon24",
    "icon36",
    "exit",
    END_SYMBOL
}
```

25.6.3.10 defined_origin_group `char* defined_origin_group[]`

Initial value:

```
= {
    "labelGridCenterX",
    "spinBoxGridCenterX",
    "labelGridCenterY",
    "spinBoxGridCenterY",
    "labelGridSizeX",
    "spinBoxGridSizeX",
    "labelGridSizeY",
    "spinBoxGridSizeY",
    END_SYMBOL
}
```

25.6.3.11 dimension_menu `char* dimension_menu[]`

Initial value:

```
= {
    "quickleader",
    END_SYMBOL
}
```

25.6.3.12 dimension_toolbar `char* dimension_toolbar[]`

Initial value:

```
= {
    "quickleader",
    END_SYMBOL
}
```

25.6.3.13 draw_menu `char* draw_menu[]`

Initial value:

```
= {
    "path",
    "polyline",
    "---",
    "arc",
    "circle",
    "ellipse",
    "line",
    "point",
    "polygon",
    "rectangle",
    "singlelinetext",
    "---",
    "dolphin",
    "heart",
    "snowflake",
    "star",
    END_SYMBOL
}
```

25.6.3.14 draw_toolbar `char* draw_toolbar[]`

Initial value:

```
= {
    "path",
    "polyline",
    "---",
    "arc",
    "circle",
    "ellipse",
    "line",
    "point",
    "polygon",
    "rectangle",
    "singlelinetext",
    "---",
    "dolphin",
    "heart",
    "snowflake",
    "star",
    END_SYMBOL
}
```

25.6.3.15 edit_menu `char* edit_menu[]`

Initial value:

```
= {
    "undo",
    "redo",
    "---",
    "cut",
    "copy",
    "paste",
    "---",
    "settingsdialog",
    END_SYMBOL
}
```

25.6.3.16 edit_toolbar `char* edit_toolbar[]`

Initial value:

```
= {
    "undo",
    "redo",
    "---",
    "cut",
    "copy",
```

```
    "paste",
    END_SYMBOL
}
```

25.6.3.17 file_menu `char* file_menu[]`

Initial value:

```
= {
    "new",
    "---",
    "open",
    ">Open &Recent",
    "---",
    "save",
    "saveas",
    "---",
    "print",
    "---",
    "windowclose",
    "---",
    "designdetails",
    "---",
    "exit",
    END_SYMBOL
}
```

25.6.3.18 file_toolbar `char* file_toolbar[]`

Initial value:

```
= {
    "new",
    "open",
    "save",
    "saveas",
    "print",
    "designdetails",
    END_SYMBOL
}
```

25.6.3.19 geometry_type_keys `char* geometry_type_keys[]`

Initial value:

```
= {
    "ARC",
    "BLOCK",
    "CIRCLE",
    "DIMALIGNED",
    "DIMANGULAR",
    "DIMARCLENGTH",
    "DIMDIAMETER",
    "DIMLEADER",
    "DIMLINEAR",
    "DIMORDINATE",
    "DIMRADIUS",
    "ELLIPSE",
    "ELLIPSEARC",
    "HATCH",
    "IMAGE",
    "INFINITELINE",
    "LINE",
    "PATH",
    "POINT",
    "POLYGON",
    "POLYLINE",
    "RAY",
    "RECTANGLE",
    "SPLINE",
    "TEXTMULTI",
    "TEXTSINGLE",
    END_SYMBOL
}
```

25.6.3.20 grid_enabled_group `char* grid_enabled_group[]`

Initial value:

```
= {
    "labelGridType", "comboBoxGridType",
    "checkBoxGridCenterOnOrigin",
```

```
    "labelGridCenterX", "spinBoxGridCenterX",
    "labelGridCenterY", "spinBoxGridCenterY",
    "labelGridSizeX", "spinBoxGridSizeX",
    "labelGridSizeY", "spinBoxGridSizeY",
    "labelGridSpacingX", "spinBoxGridSpacingX",
    "labelGridSpacingY", "spinBoxGridSpacingY",
    "labelGridSizeRadius", "spinBoxGridSizeRadius",
    "labelGridSpacingRadius", "spinBoxGridSpacingRadius",
    "labelGridSpacingAngle", "spinBoxGridSpacingAngle",
    END_SYMBOL
}
```

25.6.3.21 grid_layout `char* grid_layout[]`

Initial value:

```
= {
    "checkBoxGridLoadFromFile", "labelGridType",
    "", "comboBoxGridType",
    "checkBoxGridCenterOnOrigin",
    "labelGridCenterX", "spinBoxGridCenterX",
    "labelGridCenterY", "spinBoxGridCenterY",
    "labelGridSizeX", "spinBoxGridSizeX",
    "labelGridSizeY", "spinBoxGridSizeY",
    "labelGridSpacingX", "spinBoxGridSpacingX",
    "labelGridSpacingY", "spinBoxGridSpacingY",
    "labelGridSizeRadius", "spinBoxGridSizeRadius",
    "labelGridSpacingRadius", "spinBoxGridSpacingRadius",
    "labelGridSpacingAngle", "spinBoxGridSpacingAngle",
    END_SYMBOL
}
```

25.6.3.22 grid_load_from_file_group `char* grid_load_from_file_group[]`

Initial value:

```
= {
    "labelGridType",
    "comboBoxGridType",
    "checkBoxGridCenterOnOrigin",
    "labelGridSpacingX",
    "spinBoxGridSpacingX",
    "labelGridSpacingY",
    "spinBoxGridSpacingY",
    "labelGridSizeRadius",
    "spinBoxGridSizeRadius",
    "labelGridSpacingRadius",
    "spinBoxGridSpacingRadius",
    "labelGridSpacingAngle",
    "spinBoxGridSpacingAngle",
    END_SYMBOL
}
```

25.6.3.23 help_menu `char* help_menu[]`

Initial value:

```
= {
    "help",
    "---",
    "changelog",
    "---",
    "tipoftheday",
    "---",
    "about",
    "---",
    "whatsthis",
    END_SYMBOL
}
```

25.6.3.24 help_toolbar `char* help_toolbar[]`

Initial value:

```
= {
    "help",
    "changelog",
    "about",
    "whatsthis",
    END_SYMBOL
}
```

25.6.3.25 icon_toolbar `char* icon_toolbar[]`

Initial value:

```
= {
    "icon16",
    "icon24",
    "icon32",
    "icon48",
    "icon64",
    "icon128",
    END_SYMBOL
}
```

25.6.3.26 index_name `const char* index_name[]`

Initial value:

```
= {
    "one",
    "two",
    "three",
    "four",
    "five",
    "six",
    "seven"
}
```

25.6.3.27 index_th_name `const char* index_th_name[]`

Initial value:

```
= {
    "first",
    "second",
    "third",
    "fourth",
    "fifth",
    "sixth",
    "seventh"
}
```

25.6.3.28 inquiry_toolbar `char* inquiry_toolbar[]`

Initial value:

```
= {
    "distance",
    "locatepoint",
    END_SYMBOL
}
```

25.6.3.29 key_state `bool key_state[N_KEY_SEQUENCES]`

Initial value:

```
= {
    false, false, false, false, false,
    false, false, false, false, false,
    false, false, false, false, false,
    false, false, false, false, false,
    false, false, false, false
}
```

25.6.3.30 layer_list `char* layer_list[]`

Initial value:

```
= {
    "linetypebylayer", "0", "",
    "linetypebylayer", "1", "",
    "linetypebylayer", "2", "",
    "linetypebylayer", "3", "",
    "linetypebylayer", "4", "",
    "linetypebylayer", "5", "",
    "linetypebylayer", "6", "",
    "linetypebylayer", "7", "",
    "linetypebylayer", "8", "",
```

```
    "linetypebylayer", "9", "",
    END_SYMBOL, END_SYMBOL, END_SYMBOL
}
```

25.6.3.31 layer_toolbar `char* layer_toolbar[]`

Initial value:

```
= {
    END_SYMBOL
}
```

25.6.3.32 left_toolbar `int left_toolbar[]`

Initial value:

```
= {
    TOOLBAR_DRAW,
    TOOLBAR_MODIFY,
    TOOLBAR_BREAK,
    TERMINATOR_SYMBOL
}
```

25.6.3.33 line_type_list `char* line_type_list[]`

Initial value:

```
= {
    "linetypebylayer", "ByLayer", "",
    "linetypebyblock", "ByBlock", "",
    "linetypecontinuous", "Continuous", "",
    "linetypehidden", "Hidden", "",
    "linetypecenter", "Center", "",
    "linetypeother", "Other...", "",
    END_SYMBOL
}
```

25.6.3.34 line_weight_list `char* line_weight_list[]`

Initial value:

```
= {
    "lineweightbylayer", "ByLayer", "-2.00",
    "lineweightbyblock", "ByBlock", "-1.00",
    "lineweightdefault", "Default", "0.00",
    "lineweight01", "0.00 mm", "0.00",
    "lineweight02", "0.05 mm", "0.05",
    "lineweight03", "0.15 mm", "0.15",
    "lineweight04", "0.20 mm", "0.20",
    "lineweight05", "0.25 mm", "0.25",
    "lineweight06", "0.30 mm", "0.30",
    "lineweight07", "0.35 mm", "0.35",
    "lineweight08", "0.40 mm", "0.40",
    "lineweight09", "0.45 mm", "0.45",
    "lineweight10", "0.50 mm", "0.50",
    "lineweight11", "0.55 mm", "0.55",
    "lineweight12", "0.60 mm", "0.60",
    "lineweight13", "0.65 mm", "0.65",
    "lineweight14", "0.70 mm", "0.70",
    "lineweight15", "0.75 mm", "0.75",
    "lineweight16", "0.80 mm", "0.80",
    "lineweight17", "0.85 mm", "0.85",
    "lineweight18", "0.90 mm", "0.90",
    "lineweight19", "0.95 mm", "0.95",
    "lineweight20", "1.00 mm", "1.00",
    "lineweight21", "1.05 mm", "1.05",
    "lineweight22", "1.10 mm", "1.10",
    "lineweight23", "1.15 mm", "1.15",
    "lineweight24", "1.20 mm", "1.20",
    END_SYMBOL, END_SYMBOL, END_SYMBOL
}
```

25.6.3.35 menu_list `char* menu_list[]`

Initial value:

```
= {
    "&File",
    "&Edit",
    "&View",
```

```
    "&Draw",
    "&Tools",
    "&Modify",
    "S&andbox",
    "&Dimension",
    "&Window",
    "&Help",
    "Open &Recent",
    "&Zoom",
    "&Pan",
    END_SYMBOL
}
```

25.6.3.36 menubar_full_list `int menubar_full_list[]`

Initial value:

```
= {
    MENU_FILE,
    MENU_EDIT,
    MENU_VIEW,
    MENU_DRAW,
    MENU_TOOLS,
    MENU_MODIFY,
    MENU_DIMENSION,
    MENU_SANDBOX,
    MENU_WINDOW,
    MENU_HELP,
    TERMINATOR_SYMBOL
}
```

25.6.3.37 menubar_no_docs `int menubar_no_docs[]`

Initial value:

```
= {
    MENU_FILE,
    MENU_EDIT,
    MENU_WINDOW,
    MENU_HELP,
    TERMINATOR_SYMBOL
}
```

25.6.3.38 modify_menu `char* modify_menu[]`

Initial value:

```
= {
    "erase",
    "move",
    "rotate",
    "scale",
    END_SYMBOL
}
```

25.6.3.39 modify_toolbar `char* modify_toolbar[]`

Initial value:

```
= {
    "erase",
    "move",
    "rotate",
    "scale",
    "selectall",
    END_SYMBOL
}
```

25.6.3.40 pan_menu `char* pan_menu[]`

Initial value:

```
= {
    "panrealtime",
    "panpoint",
    "---",
    "panleft",
    "panright",
    "panup",
    "pandown",
```

```
    END_SYMBOL
}
```

25.6.3.41 patternDetailsBox `char* patternDetailsBox[]`

Initial value:

```
= {
    "labelTotalStitches", "fieldTotalStitches",
    "labelRealStitches", "fieldRealStitches",
    "labelJumpStitches", "fieldJumpStitches",
    "labelTrimStitches", "fieldTrimStitches",
    "labelTotalColors", "fieldTotalColors",
    "labelColorChanges", "fieldColorChanges",
    "labelRectLeft", "fieldRectLeft",
    "labelRectTop", "fieldRectTop",
    "labelRectRight", "fieldRectRight",
    "labelRectBottom", "fieldRectBottom",
    "labelRectWidth", "fieldRectWidth",
    "labelRectHeight", "fieldRectHeight",
    END_SYMBOL
}
```

25.6.3.42 preview_to_dialog `int preview_to_dialog[]`

Initial value:

```
= {
    GENERAL_MDI_BG_USE_LOGO,
    GENERAL_MDI_BG_USE_TEXTURE,
    GENERAL_MDI_BG_USE_COLOR,
    DISPLAY_SHOW_SCROLLBARS,
    DISPLAY_SELECTBOX_ALPHA,
    PROMPT_FONT_FAMILY,
    PROMPT_FONT_STYLE,
    PROMPT_FONT_SIZE,
    LWT_SHOW_LWT,
    LWT_REAL_RENDER,
    TERMINATOR_SYMBOL
}
```

25.6.3.43 prompt_toolbar `char* prompt_toolbar[]`

Initial value:

```
= {
    END_SYMBOL
}
```

25.6.3.44 promptColorLayout `char* promptColorLayout[]`

Initial value:

```
= {
    "labelTextColor", "buttonTextColor",
    "labelBackgroundColor", "buttonBackgroundColor",
    END_SYMBOL
}
```

25.6.3.45 properties_toolbar `char* properties_toolbar[]`

Initial value:

```
= {
    END_SYMBOL
}
```

25.6.3.46 recent_files `char recent_files[MAX_FILES][MAX_STRING_LENGTH]`

Initial value:

```
= {
    END_SYMBOL
}
```

25.6.3.47 recent_menu `char* recent_menu[]`

Initial value:

```
= {
    END_SYMBOL
}
```

25.6.3.48 rectangular_grid_group `char* rectangular_grid_group[]`

Initial value:

```
= {
    "labelGridSizeX",
    "spinBoxGridSizeX",
    "labelGridSizeY",
    "spinBoxGridSizeY",
    "labelGridSpacingX",
    "spinBoxGridSpacingX",
    "labelGridSpacingY",
    "spinBoxGridSpacingY",
    END_SYMBOL
}
```

25.6.3.49 rectangular_grid_visible_group `char* rectangular_grid_visible_group[]`

Initial value:

```
= {
    "labelGridSizeX", "spinBoxGridSizeX",
    "labelGridSizeY", "spinBoxGridSizeY",
    "labelGridSpacingX", "spinBoxGridSpacingX",
    "labelGridSpacingY", "spinBoxGridSpacingY",
    END_SYMBOL
}
```

25.6.3.50 render_hints `int render_hints[]`

Initial value:

```
= {
    DISPLAY_USE_OPENGL,
    DISPLAY_RENDERHINT_AA,
    DISPLAY_RENDERHINT_TEXT_AA,
    DISPLAY_RENDERHINT_SMOOTH_PIX,
    DISPLAY_RENDERHINT_HIGH_AA,
    DISPLAY_RENDERHINT_NONCOSMETIC,
    TERMINATOR_SYMBOL
}
```

25.6.3.51 rulerLayout `char* rulerLayout[]`

Initial value:

```
= {
    "labelRulerPixelSize", "spinboxRulerPixelSize",
    END_SYMBOL
}
```

25.6.3.52 sandbox_menu `char* sandbox_menu[]`

Initial value:

```
= {
    "simulate",
    "play",
    "pause",
    "rewind",
    "fast-forward",
    "stop",
    "export-video",
    "---",
    "qr",
    "lettering",
    "pattern",
    "design",
    END_SYMBOL
}
```

25.6.3.53 sandbox_toolbar `char* sandbox_toolbar[]`

Initial value:

```
= {
    "sandbox",
    END_SYMBOL
}
```

25.6.3.54 script_false `const ScriptValue script_false`

Initial value:

```
= {
    .r = 0.0F,
    .i = 0,
    .b = false,
    .s = "",
    .label = "false",
    .type = SCRIPT_BOOL
}
```

25.6.3.55 script_null `const ScriptValue script_null`

Initial value:

```
= {
    .r = 0.0F,
    .i = 0,
    .b = false,
    .s = "",
    .label = "NULL",
    .type = SCRIPT_NULL
}
```

25.6.3.56 script_true `const ScriptValue script_true`

Initial value:

```
= {
    .r = 0.0F,
    .i = 1,
    .b = true,
    .s = "",
    .label = "true",
    .type = SCRIPT_BOOL
}
```

25.6.3.57 selectionBoxLayout `char* selectionBoxLayout[]`

Initial value:

```
= {
    "labelCrosshairColor", "buttonCrosshairColor",
    "labelBackgroundColor", "buttonBackgroundColor",
    "labelSelectionColorCrossing", "buttonSelectionColorCrossing",
    "labelSelectionFillCrossing", "buttonSelectionFillCrossing",
    "labelSelectionColorWindow", "buttonSelectionColorWindow",
    "labelSelectionFillWindow", "buttonSelectionFillWindow",
    "labelSelectionFillAlpha", "spinboxSelectionFillAlpha",
    END_SYMBOL
}
```

25.6.3.58 settings_tab_labels `char* settings_tab_labels[]`

Initial value:

```
= {
    "General",
    "Files/Path",
    "Display",
    "Prompt",
    "Open/Save",
    "Printing",
    "Snap",
    "Grid/Ruler",
    "Ortho/Polar",
    "QuickSnap",
    "QuickTrack",
    "LineWeight",
    "Selection",
    END_SYMBOL
```

```
}
```

25.6.3.59 text_size_list `char* text_size_list[]`

Initial value:

```
= {
    "6 pt", "6", "",
    "8 pt", "8", "",
    "9 pt", "9", "",
    "10 pt", "10", "",
    "11 pt", "11", "",
    "12 pt", "12", "",
    "14 pt", "14", "",
    "18 pt", "18", "",
    "24 pt", "24", "",
    "30 pt", "30", "",
    "36 pt", "36", "",
    "48 pt", "48", "",
    "60 pt", "60", "",
    "72 pt", "72", "",
    END_SYMBOL, END_SYMBOL, END_SYMBOL
}
```

25.6.3.60 text_toolbar `char* text_toolbar[]`

Initial value:

```
= {
    END_SYMBOL
}
```

25.6.3.61 tips `char* tips[]`

Initial value:

```
= {
    "we need more tips?",
    "you can change the color of the display through settings?",
    "you can hide the scrollbars to increase the viewable area through settings?",
    "you can change the icon size for increased visibility?",
    "you can toggle the grid on and off by pressing the button in the statusbar?",
    "the grid size can be changed to match your hoop size through settings?",
    "the crosshair size is based on a percentage of your screen size?  Setting it to 100 may help you
      visually line things up better.",
    "you can pan by pressing the middle mouse button and dragging your mouse across the screen?",
    "you can open and edit multiple designs simultaneously?",
    "you that many embroidery machines support the .dst format?",
    "that you can zoom in and out using your mouse wheel?",
    "that you can use circular and isometric grids?",
    "about our command line format converter?",
    "that you can use the 'DAY' and 'NIGHT' commands to quickly switch the view colors to commonly used
      white or black?",
    "that you can quickly change the background, crosshair and grid colors using the 'RGB' command?",
    END_SYMBOL
}
```

25.6.3.62 toolbar_horizontal `int toolbar_horizontal[]`

Initial value:

```
= {
    TOOLBAR_VIEW,
    TOOLBAR_LAYER,
    TOOLBAR_PROPERTIES,
    TOOLBAR_TEXT,
    TOOLBAR_PROMPT,
    TERMINATOR_SYMBOL
}
```

25.6.3.63 toolbar_list `char* toolbar_list[]`

Initial value:

```
= {
    "File",
    "Edit",
    "View",
    "Icon",
    "Help",
```

```
    "Layer",
    "Properties",
    "Text",
    "Command Prompt",
    "Draw",
    "Modify",
    "Inquiry",
    "Dimension",
    "Sandbox",
    END_SYMBOL
}
```

25.6.3.64 toolbars_when_docs `int toolbars_when_docs[]`

Initial value:

```
= {
    TOOLBAR_VIEW,
    TOOLBAR_ICON,
    TOOLBAR_HELP,
    TOOLBAR_LAYER,
    TOOLBAR_TEXT,
    TOOLBAR_PROPERTIES,
    TOOLBAR_DRAW,
    TERMINATOR_SYMBOL
}
```

25.6.3.65 tools_menu `char* tools_menu[]`

Initial value:

```
= {
    "distance",
    "locatepoint",
    "selectall",
    END_SYMBOL
}
```

25.6.3.66 top_toolbar `int top_toolbar[]`

Initial value:

```
= {
    TOOLBAR_BREAK,
    TOOLBAR_FILE,
    TOOLBAR_EDIT,
    TOOLBAR_HELP,
    TOOLBAR_ICON,
    TOOLBAR_VIEW,
    TOOLBAR_BREAK,
    TOOLBAR_LAYER,
    TOOLBAR_PROPERTIES,
    TOOLBAR_BREAK,
    TOOLBAR_TEXT,
    TERMINATOR_SYMBOL
}
```

25.6.3.67 ui_labels `char* ui_labels[]`

Initial value:

```
= {
    "labelTextColor", "Text Color",
    "labelBackgroundColor", "Background Color",
    "labelTotalStitches", "Total Stitches:",
    "labelRealStitches", "Real Stitches:",
    "labelJumpStitches", "Jump Stitches:",
    "labelTrimStitches", "Trim Stitches:",
    "labelTotalColors", "Total Colors:",
    "labelColorChanges", "Color Changes:",
    "labelRectLeft", "Left:",
    "labelRectTop", "Top:",
    "labelRectRight", "Right:",
    "labelRectBottom", "Bottom:",
    "labelRectWidth", "Width:",
    "labelRectHeight", "Height:",
    "labelCrosshairColor", "Crosshair Color",
    "labelBackgroundColor", "Background Color",
    "labelSelectionColorCrossing", "Selection Box Color (Crossing)",
    "labelSelectionFillCrossing", "Selection Box Fill (Crossing)",
    "labelSelectionColorWindow", "Selection Box Color (Window)",
    "labelSelectionFillWindow", "Selection Box Fill (Window)",
```

```
    "labelSelectionFillAlpha", "Selection Box Fill Alpha",
    "labelRulerPixelSize", "Ruler Pixel Size",
    END_SYMBOL
}
```

25.6.3.68 view_menu `char* view_menu[]`

Initial value:

```
= {
    ">&Zoom",
    ">&Pan",
    "---",
    "day",
    "night",
    "---",
    END_SYMBOL
}
```

25.6.3.69 view_toolbar `char* view_toolbar[]`

Initial value:

```
= {
    "zoomin",
    "zoomout",
    "zoomextents",
    "---",
    "panrealtime",
    "panpoint",
    "panleft",
    "panright",
    "panup",
    "pandown",
    "---",
    "day",
    "night",
    END_SYMBOL
}
```

25.6.3.70 window_menu `char* window_menu[]`

Initial value:

```
= {
    END_SYMBOL
}
```

25.6.3.71 zoom_gb_data `WidgetData zoom_gb_data[]`

Initial value:

```
= {
    {
        .key = "labelZoomInScale",
        .label = "Zoom In Scale",
        .id = DISPLAY_ZOOMSCALE_IN
    },
    {
        .key = "labelZoomOutScale",
        .label = "Zoom Out Scale",
        .id = DISPLAY_ZOOMSCALE_OUT
    },
    {
        .key = END_SYMBOL,
        .label = END_SYMBOL,
        .id = -1
    }
}
```

25.6.3.72 zoom_layout `char* zoom_layout[]`

Initial value:

```
= {
    "labelZoomInScale", "spinboxZoomInScale",
    "labelZoomOutScale", "spinboxZoomOutScale",
    END_SYMBOL
}
```

25.6.3.73 zoom_menu `char* zoom_menu[]`

Initial value:

```
= {
    "zoomwindow",
    "zoomdynamic",
    "zoomscale",
    "---",
    "zoomcenter",
    "zoomin",
    "zoomout",
    "---",
    "zoomselected",
    "zoomall",
    "zoomextents",
    END_SYMBOL
}
```

25.7 libembroidery-thumbnailer-kde4.h

```
#ifndef LIBEMBROIDERY_THUMBNAILER_KDE4_H
#define LIBEMBROIDERY_THUMBNAILER_KDE4_H

#include <kio/thumbcreator.h>

class EmbroideryThumbnailer :  public ThumbCreator
{
public:
    EmbroideryThumbnailer();
     virtual ~EmbroideryThumbnailer();
     virtual bool create(const QString& path, int w, int h, QImage& img);
};

#endif

/* kate:  bom off; indent-mode cstyle; indent-width 4; replace-trailing-space-save on; */
```

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